Tunnelling and Underground Space Technology 142 (2023) 105430

Contents lists available at ScienceDirect

Tunnelling and Underground Space Technology incorporating Trenchless Technology Research

journal homepage: www.elsevier.com/locate/tust

3D laser point cloud-based geometric digital twin for condition assessment of large diameter pipelines

Minghao Li [a], Xin Feng [a,*], Qunfang Hu [b]

[a] Faculty of Infrastructure Engineering, Dalian University of Technology, Dalian, China
[b] Shanghai Institute of Disaster Prevention and Relief, Tongji University, Shanghai, China

ARTICLE INFO

Keywords:
Geometric digital twin
Urban pipe network
Quantitative condition assessment
3D Point cloud data
Geometric feature extraction

ABSTRACT

The increasing aging of underground pipe networks and the lack of effective inspection technologies present considerable challenges for whole life-cycle management of these infrastructures. Modern laser scanning technology offers a cost-effective and safe means to obtain dense and accurate 3D topographic data of the inner surface of pipelines. However, laser scanning point clouds contain substantial noise and outliers, and efficiently extracting valuable information for structural and functional mapping remains in its infancy. This paper presents an innovative method for fast processing point clouds data of large-diameter pipelines, enabling the accurate extraction of geometric features and efficient establishment of geometric digital twin using density-based clustering, fitting and region growing algorithm. Experimental tests were conducted to evaluate the accuracy, efficiency, and feasibility of the proposed method. The results demonstrate that the proposed approach not only robustly achieves high accuracy but also maintains high computational efficiency. Additionally, the geometric digital twin shows promise as tools for quantitatively assessing structural deformation and blockage defects.

1. Introduction

Large diameter pipelines, serving as critical urban infrastructure, are extensively employed in urban sewage and water pipeline systems amid rapid urbanization. For example, concrete pipes (CP) and reinforced concrete pipes (RCP) with diameters larger than 650 mm are commonly used in sewer systems (Sever et al., 2017; United States Environmental Protection Agency, 2023), while pre-stressed concrete cylinder pipes (PCCP) and ductile iron pipes with diameters larger than 750 mm are gaining popularity in water supply systems (Polvi et al., 2018). As urban pipelines continue to expand gradually, automating their condition assessment has become increasingly difficult due to structure deterioration under hidden underground and intricate service environments (Li et al., 2022a; Li and Feng, 2022; Huang et al., 2023).

Cross-sectional deformation, or ovality, serves as a critical performance indicator for pipelines (Loganathan et al., 2022; Sun et al., 2022), as required by Chinese code (CJJ 181–2012, 2012) and American standard (ASTM F-1216–22, 2022; DNS-OS-F101, 2012). For instance, cracks may happen when deformation reach approximately 2–3 % of the concrete pipe diameter (Bai and Bai, 2005; García and Moore, 2016; Munro et al., 2009; Xu et al., 1998). If radial deformation exceeds this threshold, the pipeline may be unable to withstand vertical loads due to reduced stiffness from pipe wall fracture, leading to larger cracking, spalling, and eventually pipeline fracture failure (Ékes, 2018; Haktanir et al., 2007), as illustrated in Fig. 1 (Tang, 2023). In addition, presence of ovality in a pipeline can significantly shorten its lifespan and impair its overall performance (Hawari et al., 2018). Modern inspection technologies for pipeline internal surface can be classified into three categories: image-based visual inspections, acoustic methods, and laser-based scanning techniques. Image-based visual inspection technologies (such as CCTV) (Elmasry et al., 2019; Haurum and Moeslund, 2020), bolstered by deep learning advancements, are quickly becoming a research hotspot for pipeline condition assessment (Tan et al., 2021). Several studies leverage CNNs, RegNetY networks, and transformer-optimized networks for automatic defect identification in sewer pipelines (Hassan et al., 2019; Li et al., 2019; Li et al., 2022b; Ma et al., 2021, 2023a). However, these methods mainly focus on classification tasks, not quantitative assessments. Various computer vision techniques, including edge detection (Kirstein et al., 2012), morphological detection (McKim and Sinha, 1999), and contour segmentation, have been employed to quantify pipeline cracks (Sinha, 2001; Fang et al., 2022), deformation (Xu et al., 1998; Kannala et al., 2008) and settled deposits

* Corresponding author.
E-mail address: fengxin@dlut.edu.cn (X. Feng).

https://doi.org/10.1016/j.tust.2023.105430
Received 22 May 2023; Received in revised form 12 September 2023; Accepted 28 September 2023
Available online 12 October 2023
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(Hawari et al., 2018). However, the limitations of 2D images only render them inappropriate for pipeline deformation assessment due to factors such as limited camera view and lens distortions (Moon et al., 2019), difficulty in aligning pipeline profile perpendicular to the pipe's axis, and poor interior illumination (Duran et al., 2003, 2007; Clemens et al., 2015; Lepot et al., 2017; Stanić et al., 2017). Additionally, acoustic methods are mainly employed when the pipeline is above 2/3 of the water level, they lack of the accuracy of laser scanning (Plihal et al., 2016). For example, geometric deformation measurement of concrete pipelines demands high accuracy and clarity, with a 1-meter diameter pipeline requiring millimeter-level precision.

The recent advancements in laser scanning provides a suitable approach for capturing the global geometry of pipelines (Zhou et al., 2022), with the aid of tripods (Olsen et al., 2010), unmanned aerial vehicle (UAV) (Chen et al., 2020), unmanned surface vessel (USV) (He et al., 2022), and robots (Wang and Yin, 2022). For example, Kawashima et al. (2014) adopted normal-based region growing and RANSAC-based cylinder surface fitting to automatically recognize piping systems. Qiu et al. (2014) utilized Gaussian sphere and RANSAC to extract pipe-runs from large-scale point clouds. Czerniawski et al. (2016) located and extracted pipe spools in cluttered point clouds, primarily using local data level curvature estimation, density-based spatial clustering of applications with noise (DBSCAN), and bag-of-features (BoF) matching. Patil et al. (2017) presented an area-based adaptive Hough transform for detecting cylinders. However, most existing automated algorithms, which utilize circles or ellipses to fit pipe cross-sections, focus on extracting and reconstructing pipeline systems rather than measuring geometric deformation. Laser scanning data generate massive point clouds, containing considerable noise and outliers (Yadav et al., 2021). In general, noise results from the measurement error of laser scanner, obeying Gaussian distribution and not exceeding 5 mm in range, while outliers arise from lighting or the reflective nature of surface in the scene (Zaman et al., 2017) and are sparsely distributed spatially. Owing to these point cloud characteristics, numerous density-based point cloud denoising methods, such as kernel density estimation (KDE) (Laefer and Truong-Hong, 2017; Truong-Hong and Lindenbergh, 2022a, 2022b; Yan and Hajjar, 2021), DBSCAN (Liu et al., 2022), voxel grid-based sampling (Pang et al., 2022), and statistical-based outlier removal (SOR), have been developed in recent years (Hou et al., 2017). Unfortunately, fast and accurately measuring the geometry of pipelines' complex curved surfaces remains challenging currently (Liu and Kleiner, 2013), as existing methods perform better for planar building surfaces (i.e., horizontal and vertical planes). Consequently, the use of laser scanning point clouds for accurate deformation measurement in the geometric digital twin of pipelines necessitates addressing the following critical issues: (1) How can noise and outliers be effectively filtered out of pipeline surface point clouds? (2) How can the feasibility of the proposed algorithm, especially concerning accuracy and time consumption evaluation, be effectively verified?

To address the aforementioned challenges, this paper proposes a novel method for the fast and accurate extraction of pipeline surfaces and geometric digital twin modeling using density-based fitting, clustering and region growing. The proposed algorithm comprises three components: (1) point cloud preprocessing, (2) geometric feature extraction utilizing least-median-of-squares (LMedS) fitting and DBSCAN clustering, and (3) 3D geometric digital twin modeling through slice region growing. A series of laboratory experiments are designed to assess the accuracy and computational time of the proposed method, thereby verifying its feasibility for quantitatively assessing structural deformation as well as blockage defects in the pipelines.

The remainder of this paper is structured as follows. Section 2 details the methodology and overall framework. In Section 3, the proposed approach is validated, and its performance is compared with existing methods. Lastly, Section 4 presents the conclusions.

2. Methodology

The underground pipeline system has been deemed one of the most deteriorated infrastructures, with a "D+" rating evaluated by ASCE (2021 Infrastructure Report Card, 2021), due to harsh service environments and hidden degradations such as deformation, concrete spalling, scaling, and blockages (Kirkham et al., 2000). Laser scanning, an efficient technique for accurately capturing object geometries, enables the acquisition of 3D spatial data mapping the actual deformation and defects of underground pipelines in hidden and dark environments that are inaccessible to humans. Nevertheless, the point cloud data typically contains substantial noise and outliers, making it unsuitable for direct use in computing deformation and defect volumes. The overarching goal

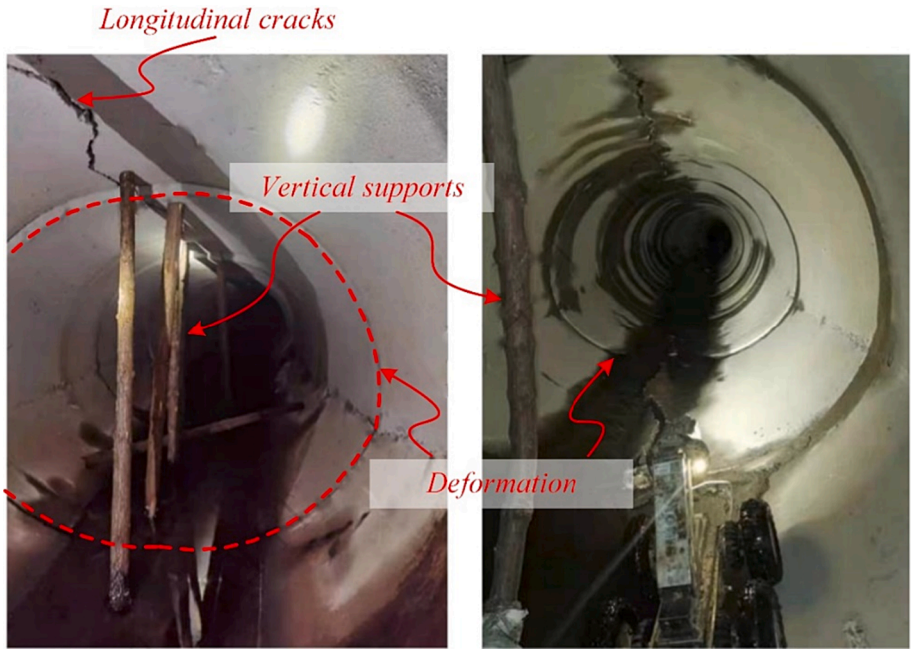


Fig. 1. Deformation damage of an urban rain and sewer reinforced concrete pipe (Tang, 2023).

of this paper is to develop a novel methodology, transforming raw point cloud data into a geometric digital twin that can fast and accurately assess pipeline deformation and defects. As illustrated in Fig. 2, the proposed methodology consists of three steps: (1) To circumvent the exhausted computations associated with 3D data processing, cluttered 3D points are initially corrected and segmented into 2D slice-regions along the z-axis; (2) Employing the knowledge of scanner measurement errors and pipeline surface characteristics, LMedS and DBSCAN-based fitting and clustering algorithms are proposed to extract geometric features of the slices; and (3) Through quantitative comparison of the similarities between adjacent slice regions, density-based region growing (DRG) is proposed to significantly reduce inefficient computations and extract 3D geometric features. The proposed method pertains to extracting the geometric features and establishing the geometric digital twin to quantitatively assess the deformation and defects for large diameter pipelines.

2.1. 3D point-to-2D region

Since laser scanning point clouds are typically massive, it can be time-consuming to analyze each point in a 3D manner. In this study, because 2D processing is simpler and more efficient than 3D processing, the 3D point cloud is segmented into several 2D slice-regions for data processing (i.e., 3D point-to-2D region), which are perpendicular to the pipe axis direction. However, since a pipeline can have an arbitrary orientation in geo-space, it is necessary to align the pipeline axis parallel to the coordinate axis z. Consequently, the 3D point-to-2D region methodology consists of two sub-steps: pipeline axis alignment and slice region subdivision.

2.1.1. Pipeline axis alignment

The objective of this section is to align pipe axis perpendicular to the *xoy* plane, ensuring that the slices can be more representative of the pipe cross-section. In order to reduce the computational complexity and time, voxel-based down-sampling is employed to simplify the raw point cloud ($\boldsymbol{P}_{raw} = \{p_i = (x_i, y_i, z_i) \in R^3, i = [1, N_{raw}]\}$) into a smaller number of data points ($\boldsymbol{P}_{ds} = \left\{p_{ds,i} = \left(x_{ds,i}, y_{ds,i}, z_{ds,i}\right) \in R^3, i = [1, N_{ds}]\right\}$). Subsequently, normal vectors of the point cloud $\boldsymbol{P}_{ds}$ are estimated and projected onto a unit sphere (Gaussian sphere) surface (Czerniawski et al., 2016; Kawashima et al., 2014; Patil et al., 2017; Qiu et al., 2014), with the normal vectors unitized as $\boldsymbol{P}_n = \left\{p_{n,i} = \left(x_{n,i}, y_{n,i}, z_{n,i}\right) \in R^3, i = [1, N_{ds}]\right\}$. A plane is fitted using the RANSAC method, which has the normal vector $v_n = (\alpha, \beta, \gamma)$ in the same direction as the pipeline axis. Consequently, the Euler angles in the rotation matrix $\boldsymbol{R}$ along the x, y and z axes are $arctan(\beta/\gamma)$, $-arctan(\alpha/\gamma)$ and 0, respectively. After rotating the raw point cloud $\boldsymbol{P}_{raw}$ by the rotation matrix, the axis of the rotated point cloud ($\boldsymbol{P}_z = \left\{p_{z,i} = \boldsymbol{R}\cdot(x_i, y_i, z_i)^T \in R^3, i = [1, N_{raw}]\right\}$) is considered to be parallel to the z-axis. This alignment supports cropping of the pipe point cloud into regularly sliced regions using the axes of the global coordinate system in following steps, thus reducing the computational complexity. The workflow of pipeline axis alignment is illustrated in Fig. 3.

2.1.2. Slice region subdivision

The cross-sections of a pipeline along the longitudinal direction are predominantly circular. To leverage these geometric properties and decrease the complexity of the entire point cloud, slicing is employed to extract geometric features faster in a 2D manner and to recursively construct the 3D geometric digital twin of a pipeline based on the similarity between slices. The initial 3D boundaries $(x_{b_min}, x_{b_max}, y_{b_min}, y_{b_max}, z_{b_min}, z_{b_max})$ of the corrected point cloud ($\boldsymbol{P}_z = \left\{p_{z,i} = \boldsymbol{R}\cdot(x_i, y_i, z_i)^T \in R^3, i = [1, N_{raw}]\right\}$) are first estimated. Define Δt as the division step size, which depends on a specific required level of detail for the structure (e.g., the size of defects such as spalling and blockage). Subdivide the point cloud $\boldsymbol{P}_z$ into 2D regions ($\boldsymbol{S}_i = \left\{s_{i_j} = \left(x_{i_j}, y_{i_j}\right) \in R^2, j = [1, N_r]\right\}, i = [1, N_{slice}]$ where $z_{b_min} + (i-1)\Delta t \le z \le z_{b_min} + i\Delta t$) along the z-axis using a bounding box, with N_r representing the number of the points within the $\boldsymbol{S}_i$. Additionally, to extract slices with complete geometric information, following the approach introduced by Truong-Hong and Lindenbergh (2022a, 2022b), if the slice contains fewer points than a predefined threshold ($N_r < n_{r_min}$), each slice is designated as "*empty*"; otherwise, it is classified as "*full*". Only full slices are considered in subsequent data processing steps. Through slicing, the complex 3D point cloud is transformed into simpler 2D planar data points $\boldsymbol{S}_i$, including its z-axis coordinate information $z_{c,i} = z_{b_min} + (i - 0.5)\Delta t$. Note that the number of slices is $N_{slice} = \text{ceil}((z_{b_max} - z_{b_min})/\Delta t)$, where ceil(*) indicates rounding up to an integer. The workflow of slice region subdivision is depicted in Fig. 4.

2.2. Geometric feature extraction

Given that the 3D point cloud of the pipeline has been decomposed into 2D slices in the *xoy* plane, the geometry represented by the set of 2D points closely resembles a circle or an ellipse. Most existing methods directly simplified it to circle (Liu et al., 2022) or ellipse (Kannala et al., 2008; Xu et al., 1998). However, research has shown that the pipeline deformation under pipe-soil interaction takes on irregular shapes (Haktanir et al., 2007), as shown in Fig. 1. Cracks may happen when the

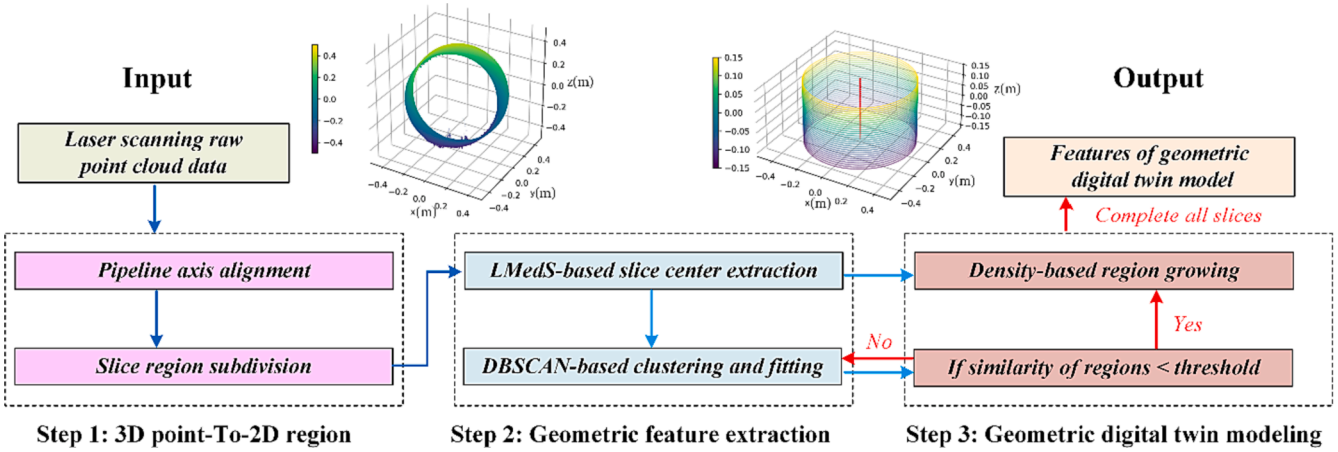


Fig. 2. Workflow for deformation measurement and geometric digital twin modeling of large diameter pipelines.

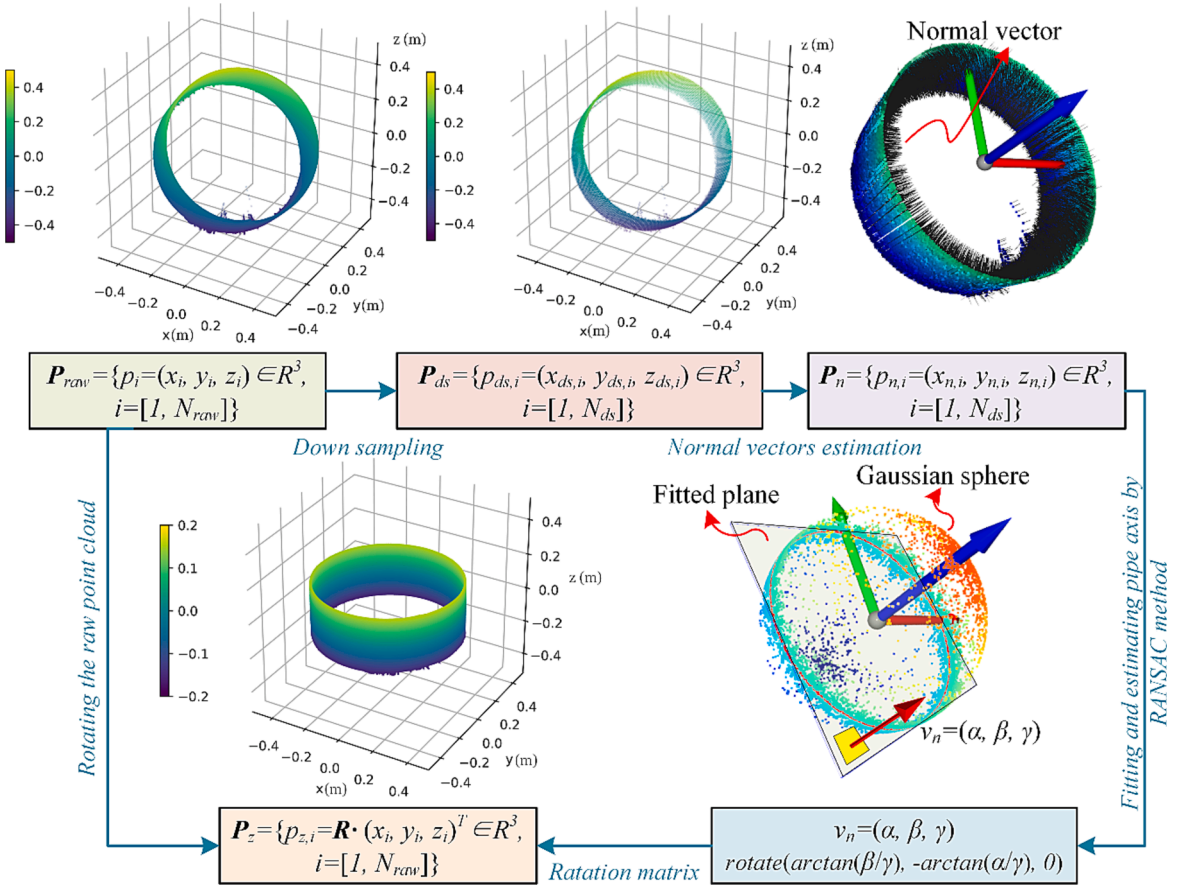


Fig. 3. Workflow of axis alignment.

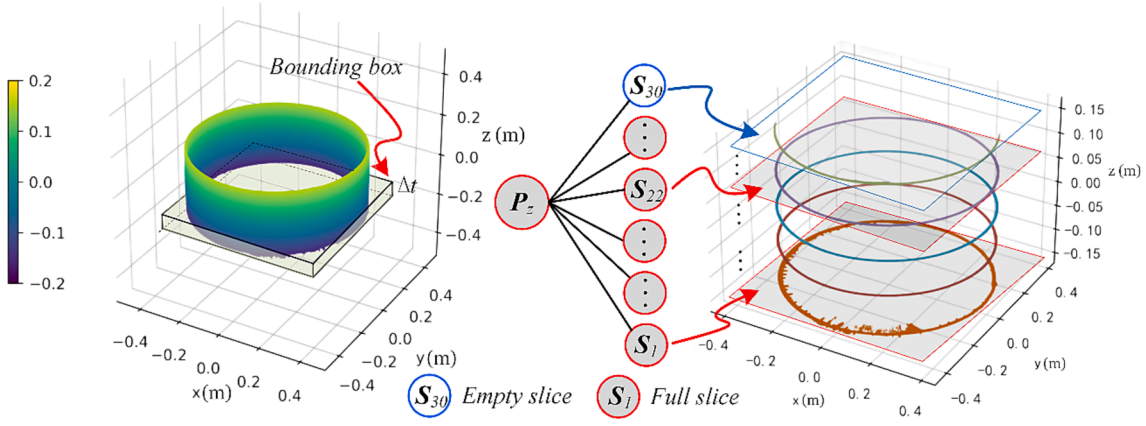


Fig. 4. Workflow of slice region subdivision.

ovality of the pipeline reaches 2 %, that is to say, for a concrete pipe with a diameter of 800 mm, its threshold of radial deformation will be 2 cm, so the accuracy requirement should be millimeter level. The accurate representation of the cross-section using a circle or ellipse is not feasible due to the non-uniform radial deformation commonly occurred in actual pipelines. Moreover, the precise expression and measurement of surface defects, such as blockage, spalling, or rupture, in pipeline geometry, pose a challenge. To address this issue, recent studies have demonstrated the utilization of density-based data processing methods, including DBSCAN (Czerniawski et al., 2016), which have become increasingly popular due to their efficient noise removal performance. Fig. 5 illustrates the point cloud data of a slice. The point cloud can be segmented into three categories: (1) internal and external outliers, typically located away from the main point cloud, resulting from illumination and anomalous reflections in the scene; (2) the profile line of the pipe, situated at the highest density of the main point cloud, caused by the reflection of the inner pipe surface; (3) noise, primarily located within the main point cloud, obtained by the measurement errors of the acquisition equipment and sensors. In light of the above distribution characteristics of a point cloud slice, this paper proposes a cell-based method to extract geometric features of pipe cross-sections using DBSCAN. The detailed steps are outlined below.

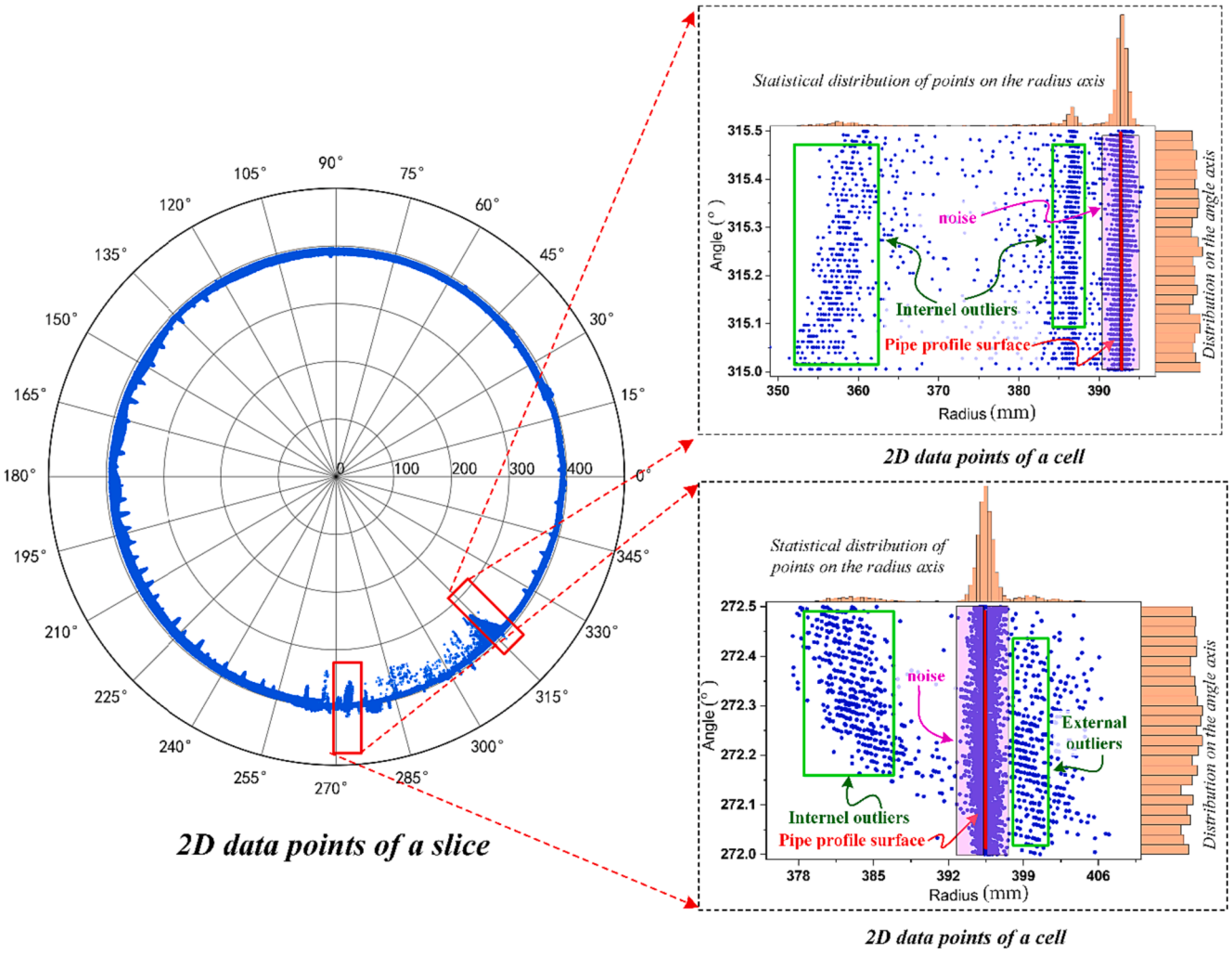


Fig. 5. Classification and distribution of laser scanning point cloud for a pipeline.

2.2.1. Extracting and calibrating slice center based on LMedS

The geometric center of a slice typically deviates from the center of the coordinate system due to the scanners' arbitrary position during measurement. This discrepancy in radius (i.e., the distance from a cell to the coordinate center) can lead to significant differences in the number of points within each cell, consequently increasing the deviation of feature estimation. Therefore, the LMedS method is proposed to extract and calibrate the slice center (Fig. 6). To minimize computation time, voxel-based down-sampling of the data ($\boldsymbol{S}_{ds,i} = \left\{ s_{ds,i_j} = \left(x_{ds,i_j}, y_{ds,i_j} \right) \in R^2, j = [1, N_{rds}] \right\}$) in the i-th slice ($\boldsymbol{S}_i = \left\{ s_{i_j} = \left(x_{i_j}, y_{i_j} \right) \in R^2, j = [1, N_r] \right\}$) is initially performed. To find the slice center, an elliptic function with five parameters is defined by Eq. (1):

$$Q(x,y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, \quad with\ 4A \cdot C - B^2 > 0 \tag{1}$$

By randomly selecting 5 points from the downsampled points $\boldsymbol{S}_{ds,i}$, Eq. (1) is iteratively computed N_m times to produce N_m ellipse models. The estimates that yield the smallest median value of squared residuals value, computed for the N_m ellipse models, are given in Eqs (2) and (3).

$$r_{i_j_k} = 2a_{i_k} - \left(\sqrt{\left(x_{ds,i_j} - x_{f1,i_k} \right)^2 + \left(y_{ds,i_j} - y_{f1,i_k} \right)^2} + \sqrt{\left(x_{ds,i_j} - x_{f2,i_k} \right)^2 + \left(y_{ds,i_j} - y_{f2,i_k} \right)^2} \right) \tag{2}$$

$$r^2_{LMedS,i} = \min_k \underset{j}{med} r^2_{i_j_k}, \quad i = [1, N_{slice}];\ j = [1, N_{rds}]\ and\ k = [1, N_m] \tag{3}$$

where $r_{i_j_k}$ denotes the residual between the k-th ellipse model and the j-th point (x_{ds,i_j}, y_{ds,i_j}) of the point set $\boldsymbol{S}_{ds,i}$ in the i-th slice; a_{i_k} represents the semi-major axis of the k-th ellipse model in the i-th slice; (x_{f1,i_k}, y_{f1,i_k}) and (x_{f2,i_k}, y_{f2,i_k}) are the two focal points of the k-th ellipse in the i-th slice; and $r_{LMedS,i}$ denotes the least value for the median of squared residuals $r_{i_j_k}$ in the i-th slice.

Subsequently, according to the review by Meer et al. (1991), in the presence of Gaussian noise, Eq. (4) and (5) are adopted to adaptively and robustly classify the inliers and outliers of the point set $\boldsymbol{S}_{ds,i}$. As shown in Fig. 6, once the inliers and outliers are determined, the optimal ellipse and its central coordinate $(x_{c,i}, y_{c,i})$ are computed using the least square method (Eq. (6)). Finally, the data points in the i-th slice are translated to $\boldsymbol{S}_{cal,i} = \left\{ s_{cal,i_j} = \left(x_{i_j} - x_{c,i}, y_{i_j} - y_{c,i} \right) \in R^2, j = [1, N_r] \right\}$.

$$\widehat{\sigma}_i = 1.4826 \left(1 + \frac{5}{N_{rds} - 5} \right) \underset{j}{\text{med}} \sqrt{\left(|r_{i_j}| - \sqrt{r^2_{LMedS,i}} \right)^2} \tag{4}$$

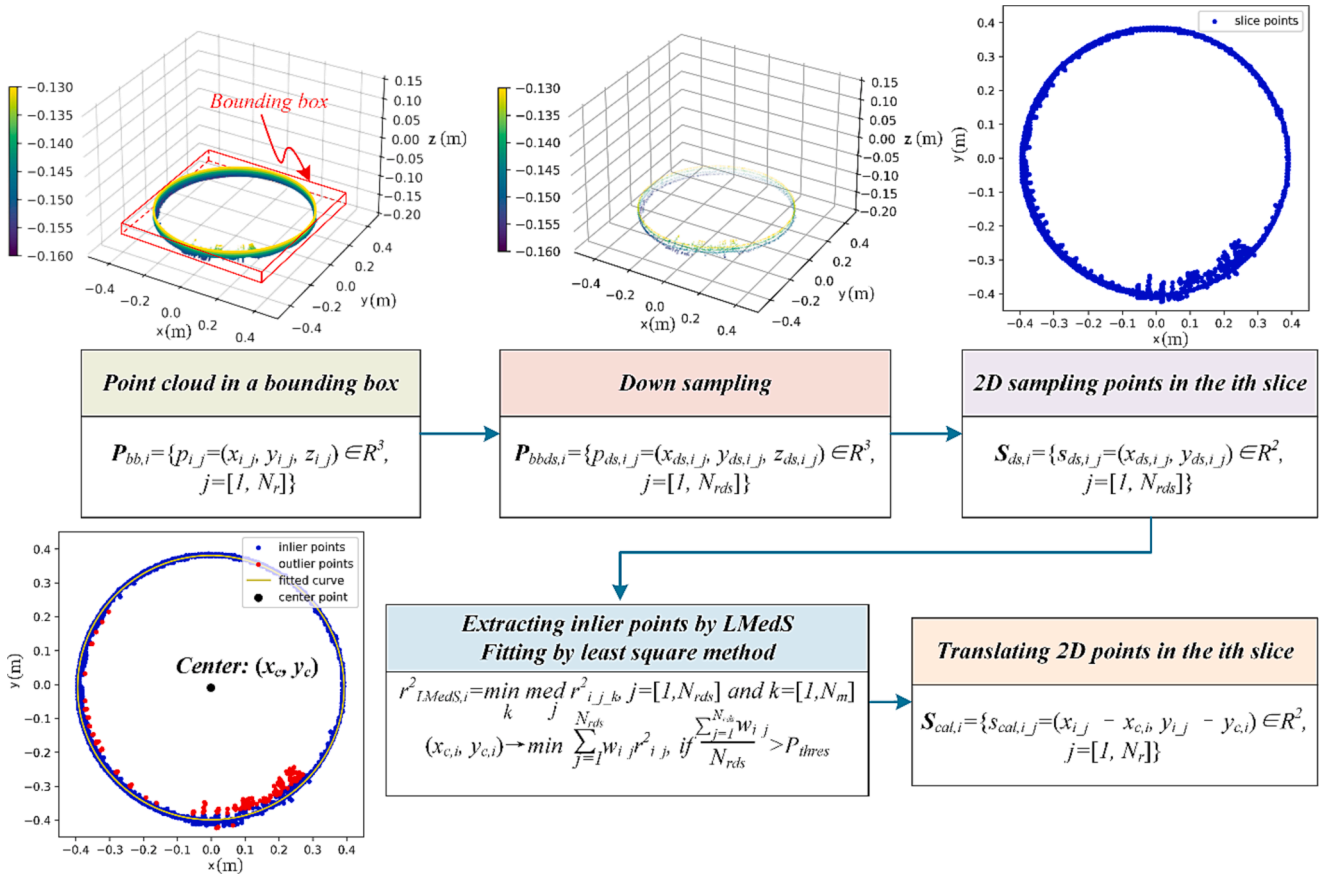


Fig. 6. Extracting and calibrating slice center based on LMedS.

$$w_{i_j} = \begin{cases} 1, if |r_{i_j}| \le 2.5\hat{\sigma} \\ 0, if |r_{i_j}| \rangle 2.5\hat{\sigma} \end{cases} \tag{5}$$

$$(x_{c,i}, y_{c,i}) \rightarrow min \sum_{j=1}^{N_{rds}} w_{i_j} r^2_{i_j}, if \frac{\sum_{j=1}^{N_{rds}} w_{i_j}}{N_{rds}} > P_{thres} \tag{6}$$

where w_{i_j} represents the weight, and if the j-th point of the i-th slice is an inlier ($w_{i_j} = 1$), it will belong to the optimal ellipse model; otherwise, the outliers ($w_{i_j} = 0$) are not considered further. P_{thres} denotes the probability threshold of inliers. If the inlier probability is below the threshold P_{thres}, the LMedS procedure needs to be re-executed until the threshold is satisfied. Note that the proportion of inliers in dataset should not be less than P_{thres}, otherwise the algorithm may not converge.

2.2.2. Clustering and fitting based on DBSCAN and parameter estimation

As previously mentioned, to accurately extract the irregular deformation of the pipeline and remove the outliers and noise from the point cloud, the slice must be subdivided into cells. Initially, the calibrated 2D points of the i-th slice ($S_{cal,i} = \left\{ s_{cal,i_j} = \left(x_{i_j} - x_{c,i}, y_{i_j} - y_{c,i} \right) \in R^2, j = [1, N_r] \right\}$) are converted to polar coordinate (θ, ρ) using Eq. (7), as this represents more suitable than Cartesian coordinate (x, y) for dividing cells on a curved surface (Ebrahimi and Jalali, 2022).

$$\begin{cases} \theta_{i_j} = \arctan\left(\dfrac{y_{i_j} - y_{c,i}}{x_{i_j} - x_{c,i}} \right) \cdot 180 \Big/ \pi \\ \rho_{i_j} = \sqrt{\left(x_{i_j} - x_{c,i} \right)^2 + \left(y_{i_j} - y_{c,i} \right)^2} \end{cases}, \; j = [1, N_r] \tag{7}$$

where θ_{i_j} denotes the angle value of the j-th point in the i-th slice; ρ_{i_j} represents the distance (radius) from the j-th point of the i-th slice to the coordinate origin (or geometric center).

Next, cells are divided based on the angle values and their ranges, which depend on the measurement standard deviation of the scanner and the design value of the pipe diameter, as expressed in Eq. (8). This actually provides a minimum range criterion for cell division, as it is meaningless to have a length less than the measurement accuracy of the scanner.

$$\begin{aligned} \boldsymbol{C}_{i_k} &= \left\{ c_{i_j} = \left(\theta_{i_j}, \rho_{i_j} \right) \in \boldsymbol{S}_{cal,i}, j = [1, N_r] \right\}, \\ & i = [1, N_{slice}] \; and \; k = [1, 360/\Delta\theta] \\ where & (k-1)\Delta\theta \le \theta_{i_j} < k\Delta\theta, \; \Delta\theta \ge 360e/(d\pi) \end{aligned} \tag{8}$$

where $\boldsymbol{C}_{i_k}$ represents the k-th cell of the i-th slice; $\Delta\theta$ denotes the cell division interval, which defines the resolution of surface geometry features; e is the measurement standard deviation of the scanner; and d refers to the design diameter of the pipeline.

Given that the outliers are generally sparsely distributed spatially, whereas most noise is Gaussian distributed along the pipe profile (Fig. 5), the DBSCAN algorithm is employed to remove outliers and partial noise, while a Gaussian fit is utilized to eliminate the remaining primary noise. DBSCAN is an unsupervised machine learning clustering algorithm that defines, by the parameters (ϵ, *MinPts*), that the neighborhood of each point of a cluster must contain at least the minimum number of points, also referred to as clustering by density. Here, ϵ defines the neighborhood distance threshold (radius) around a data point, and *MinPts* describes the minimum number of neighbors (data points) within ϵ radius. The parameters (ϵ, *MinPts*) significantly impact clustering results. As shown in Fig. 7, a common approach involves using the sorted k-distance plot to compute the optimal ϵ (i.e., the threshold between noise and other clusters) (Liu et al., 2022; Sander et al., 1998), and the *MinPts* should be higher than the number of data point

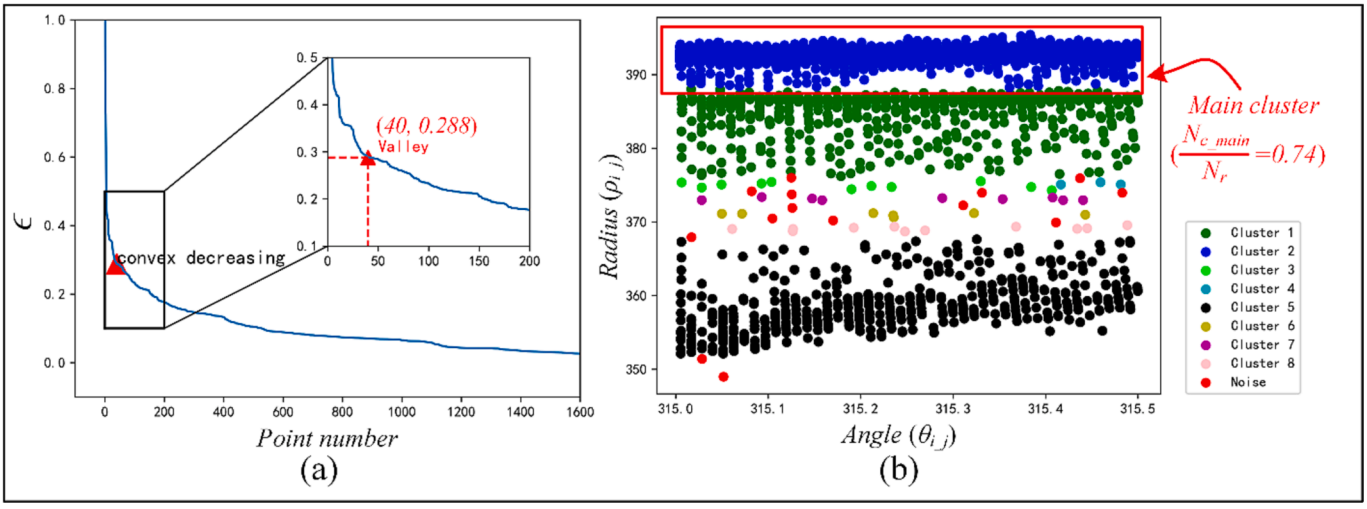


Fig. 7. Coarse extraction of pipe profile. (a) the sorted k-distance plot of the cell; (b) the results of point cloud clustering using DBSCAN.

dimensions. Notably, this step aims to segment the main point clouds (pipe profile) instead of accurately clustering each outlier. Therefore, DBSCAN can accomplish coarse extraction of the pipe profile.

After the coarse extraction of the main cluster ($\boldsymbol{C}_{main,i_k} = \{c_{main,i_l} = (\theta_{i_l}, \rho_{i_l}), l = [1, N_{c_main}]\} \subseteq \boldsymbol{C}_{i_k}$, $i = [1, N_{slice}]$ *and* $k = [1, 360/\Delta\theta]$, where $\boldsymbol{C}_{main,i_k}$ is the main cluster of the k-th cell of the i-th slice and its number of points is N_{c_main}, as shown in Fig. 6), the remaining noise, which is assumed to follow a Gaussian distribution due to scanner measurement standard deviation, is finely filtered by fitting $N\,(mean\{\rho_{i_l}\}, std\{\rho_{i_l}\})$ to obtain a geometric feature $f_{i_k} = ((k-1/2)\Delta\theta, mean\{\rho_{i_l}\})$ of the k-th cell (Fig. 8). Finally, all geometric features $\boldsymbol{F}_i = \{f_{i_k}, k = [1, 360/\Delta\theta]\}, i = [1, N_{slice}]$ of the i-th slice can be computed based on DBSCAN clustering and Gaussian distribution fitting, as demonstrated in Fig. 8.

The workflow for extracting the geometric features of the i-th slice is presented in Algorithm 1. The proposed method provides a fast solution for the accurate measurement and computation of irregular pipeline deformation. Meanwhile, it is worth noting that compared with the KDE method, which needs consideration of the contribution of each point in the cell to the density distribution, the proposed method first roughly screens out the main cluster (i.e., the pipe profile point cloud containing noise) using DBSCAN, and then extracts features finely using a Gaussian fit that saves a significant amount of computational time.

Algorithm 1: Extracting geometric features of i-th slice

Input: data points $S_{cal,i} = \left\{s_{cal,i_j} = \left(x_{i_j} - x_{c,i}, y_{i_j} - y_{c,i}\right) \in R^2, j = [1, N_r]\right\}$ of the i-th slice
Output: $\boldsymbol{F}_i = \{f_{i_k}, k = [1, 360/\Delta\theta]\}$
Begin
1. $(x_{i_j} - x_{c,i}, y_{i_j} - y_{c,i}) \rightarrow (\theta_{i_j}, \rho_{i_j})$
2. $k = 1$ and $\Delta\theta \geq 360e/(d\pi)$;
3. while $k \leq 360/\Delta\theta$:
  Dividing cell $c_{i_j} = (\theta_{i_j}, \rho_{i_j}) \in \boldsymbol{S}_{cal,i}$ according to angle, where $(k-1)\Delta\theta \leq \theta_{i_j} < k\Delta\theta$;
  ϵ is the "valley" of the sorted k-distance plot, then DBSCAN(eps=ϵ, minPts = 4);
  $\boldsymbol{C}_{main,i_k} = \{c_{main,i_l} = (\theta_{i_l}, \rho_{i_l}), l = [1, N_{c_main}]\}$;
  $\mu = \text{mean}\{\rho_{i_l}\}$ and $\sigma = \text{std}\{\rho_{i_l}\}$;
  $f_{i_k} = ((k-0.5)\Delta\theta, \mu)$;
  k++;
4. return $\boldsymbol{F}_i = \{f_{i_k}, k = [1, 360/\Delta\theta]\}$

2.3. Geometric digital twin modeling

Geometric digital twin, as a component of the digital twin models, should accurately map the actual deformation of the pipeline (Li et al., 2022a). However, most existing 3D pipeline modeling methods (Kannala et al., 2008; Kawashima et al., 2014; Patil et al., 2017) depend on

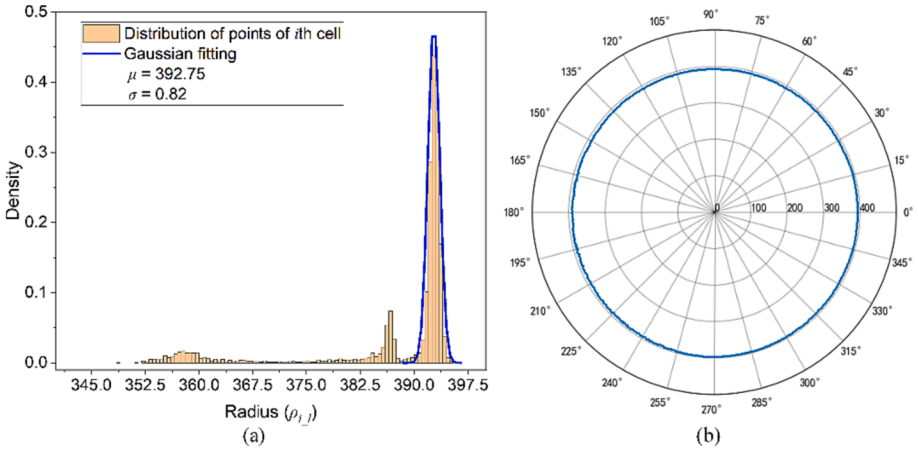


Fig. 8. Extraction of geometric features: (a) a geometric f_{i_k} of the k-th cell; (b) all geometric features $\boldsymbol{F}_i$ of the i-th slice.

design drawings or elliptical section fitting, which hardly capture pipeline deformation or defects effectively. Additionally, the tens of millions of scanned point clouds make surface reconstruction both challenging and time-consuming. Recently, some voxel-based surface reconstruction methods (Pang et al., 2022; Su et al., 2016) have been proposed, but the reconstructed polygonal mesh models may lead to inaccurate measurements of deformation and defects due to insufficient noise filtering. This section proposes a density-based region growing algorithm to improve the computational efficiency of geometric digital twin modeling, based on the accurately extracted geometry features in Section 2.2. Subsequently, an automated workflow is presented for geometric digital twin-informed quantitative condition assessment of pipelines.

2.3.1. Density-based region growing (DRG)

According to Section 2.2, the proposed Algorithm 1 can accurately and robustly extract the geometric features that represent pipeline deformation. However, the surface of neighboring regions is continuous and homogeneous. To further reduce computation time and increase efficiency (Poux et al., 2022), a DRG method is proposed to merge two highly similar adjacent regions. All slices $\boldsymbol{S}_i = \left\{ s_{i_j} = \left(x_{i_j}, y_{i_j} \right) \in R^2, j = [1, N_r] \right\}, i = [1, N_{slice}]$ initially perform the slice center calibration step in Section 2.2.1. Subsequently, the point cloud of the i-th slices is projected onto the x-axis and y-axis, and the point density distribution along the x-axis $\boldsymbol{D}_{x,i}$ and along the y-axis $\boldsymbol{D}_{y,i}$ are computed by Eqs. (9) and (10), respectively.

$$\boldsymbol{D}_{x,i} = \left\{ d_{x,i_n}, where x_{min} + (n-1)\Delta x \le x \le x_{min} + n\Delta x, n = [1, (x_{max} - x_{min})/\Delta x] \right\}, i = [1, N_{slice}] \tag{9}$$

$$\boldsymbol{D}_{y,i} = \left\{ d_{y,i_n}, where y_{min} + (n-1)\Delta y \le y \le y_{min} + n\Delta y, n = [1, (y_{max} - y_{min})/\Delta y] \right\}, i = [1, N_{slice}] \tag{10}$$

($\Delta x \ge e$ and $\Delta y \ge e$).

where d_{x,i_n} or d_{y,i_n} is the discrete probability density of the points in the interval $[x_{min} + (n-1)\Delta x, x_{min} + n\Delta x]$ or $[y_{min} + (n-1)\Delta y, y_{min} + n\Delta y]$; x_{min} or y_{min} and x_{max} or y_{max} are predefined minimum and maximum bounds of x or y-axis, respectively; Δx and Δy are the size of the grouped intervals along the x and y-axis, which should be larger than the scanner measurement standard deviation e.

To measure the similarity between the probability distribution of the i-th slice and the j-th slice, the Bhattacharyya coefficient (Bhattacharyya, 1943) is adopted, as shown in Eqs. (11) and (12):

$$\psi(\boldsymbol{D}_{x,i}, \boldsymbol{D}_{x,j}) = \sum_{n=1}^{(x_{max} - x_{min})/\Delta x} \sqrt{d_{x,i_n} \cdot d_{x,j_n}} \tag{11}$$

$$\psi(\boldsymbol{D}_{y,i}, \boldsymbol{D}_{y,j}) = \sum_{n=1}^{(y_{max} - y_{min})/\Delta y} \sqrt{d_{y,i_n} \cdot d_{y,j_n}} \tag{12}$$

where $\psi(\boldsymbol{D}_{x,i}, \boldsymbol{D}_{x,j})$ and $\psi(\boldsymbol{D}_{y,i}, \boldsymbol{D}_{y,j})$ denote the similarities between the i-th slice and j-th slice along the x- and y-axis, respectively.

Assuming that the adjacent region merging starts from slice $\boldsymbol{S}_i$ and checking whether the adjacent region $\boldsymbol{S}_j$ can be merged. If the density distribution of $\boldsymbol{S}_i$ and $\boldsymbol{S}_j$ exhibit high similarity along the x- and y-axis (Fig. 9a), they are considered to share the same geometric features, i.e., $\boldsymbol{F}_i = \boldsymbol{F}_j$. If overall deformation or local defect are present in $\boldsymbol{S}_i$ and $\boldsymbol{S}_j$, there will be differences in the density distribution (Fig. 9b); hence, $\boldsymbol{F}_i \neq \boldsymbol{F}_j$, and their geometric features are computed independently. In this implementation, sliced regions $\boldsymbol{S}_i$ and $\boldsymbol{S}_j$ are merged if their similarities are larger than a predefined similarity threshold ($\psi_{thres} = 0.995$) according to Eqs. (11) and (12). This value was selected empirically in this study, comparing also Fig. 9a and b. Notably, this value can be adjusted based on requirements of modeling accuracy and data quality.

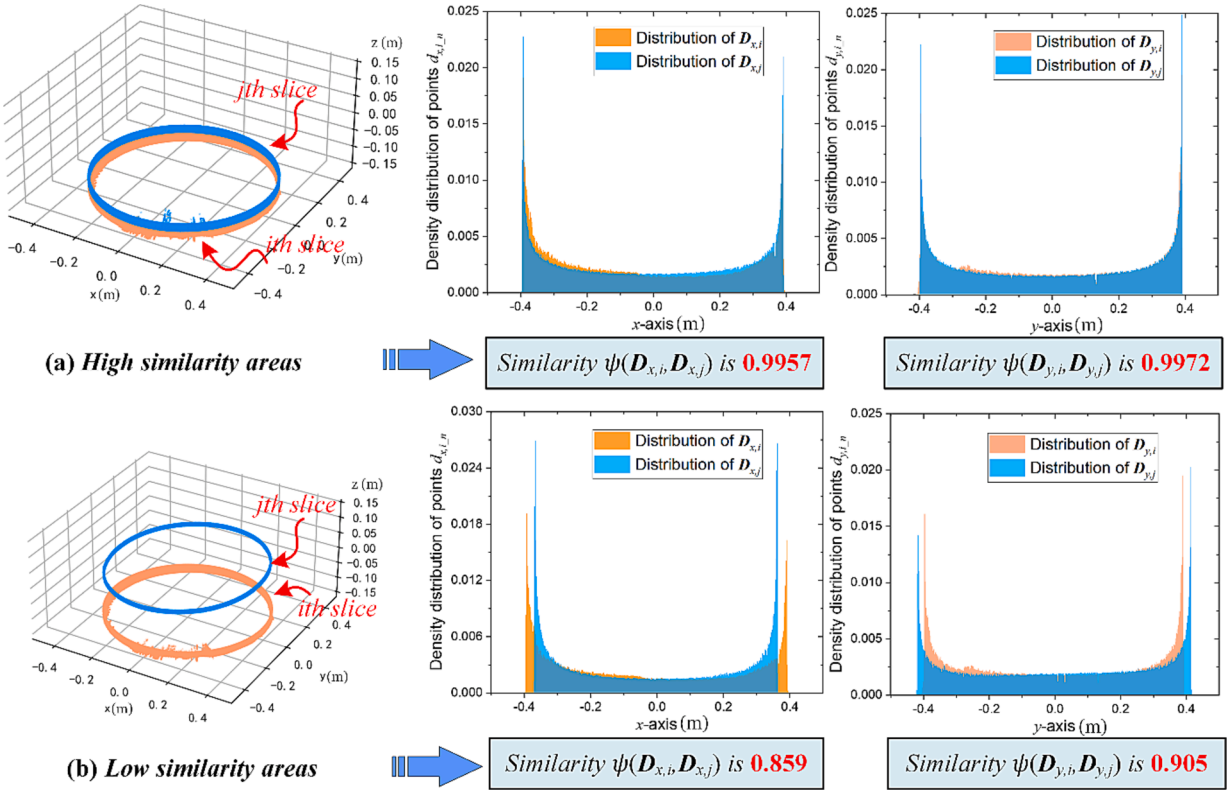
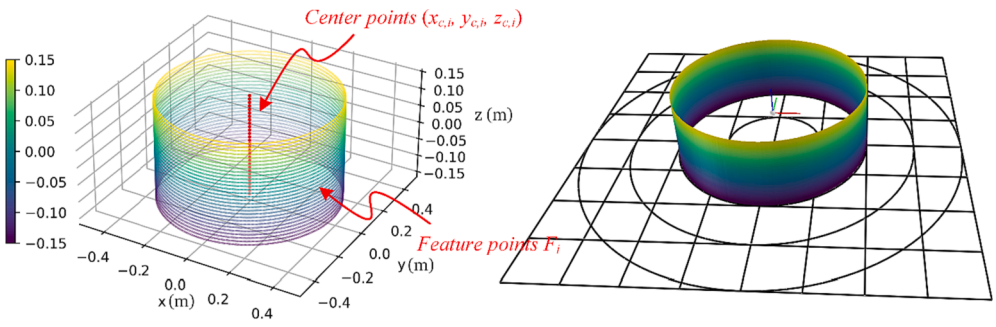


Fig. 9. Density-based region growing (DRG) method using similarity.



(a) Features of geometric digital twin (b) Geometric digital twin modeling for a pipeline

Fig. 10. 3D geometric digital twin using DRG.

2.3.2. Geometric digital twin of pipelines

The workflow for extracting geometric features of the digital twin is shown in Algorithm 2 and Fig. 10a. In contrast to existing pipeline point cloud processing approaches, the proposed algorithm remains information on deformations and defects through the distributed features. The 3D pipe surface is finely and accurately described by features carefully filtered for outliers and noise based on point cloud density, achieving an accurate mapping of deformations and defects. In the use of the modeling approach in reference (Li et al., 2023), the mesh of the geometric digital twin can be modelled (Fig. 10b) through the vertices and triangles represented by Eqs. (13) and (14).

$$Vertex\left[i\cdot\left(\frac{360}{\Delta\theta}\right)+j\right]=\left(f_{i_j}[\rho]\cdot\cos\left(f_{i_j}[\theta]\right)+x_{c,i},f_{i_j}[\rho]\cdot\sin\left(f_{i_j}[\theta]\right)+y_{c,i},z_{c,i}\right)$$
$$(i=[1,N_{slice}]\ and\ j=[1,360/\Delta\theta])\tag{13}$$

$$Traingles(v_1,v_2,v_3)=\begin{cases}\left(i\cdot\frac{360}{\Delta\theta}+j,i\cdot\frac{360}{\Delta\theta}+j+1,i\cdot\frac{360}{\Delta\theta}+j+\frac{360}{\Delta\theta}\right),\\ where\,i=[1,N_{slice}-1],j=\left[1,\frac{360}{\Delta\theta}-1\right];\\ \left(i\cdot\frac{360}{\Delta\theta}+j+1,i\cdot\frac{360}{\Delta\theta}+j+\frac{360}{\Delta\theta}+1,i\cdot\frac{360}{\Delta\theta}+j+\frac{360}{\Delta\theta}\right),\\ where\,i=[1,N_{slice}-1],j=\left[1,\frac{360}{\Delta\theta}-1\right];\\ \left(i\cdot\frac{360}{\Delta\theta}+j,i\cdot\frac{360}{\Delta\theta}+j-\frac{360}{\Delta\theta}+1,i\cdot\frac{360}{\Delta\theta}+j+\frac{360}{\Delta\theta}-1\right),\\ where\,i=[1,N_{slice}-1],j=\frac{360}{\Delta\theta};\\ \left(i\cdot\frac{360}{\Delta\theta}+j-\frac{360}{\Delta\theta}+1,i\cdot\frac{360}{\Delta\theta}+j+1,i\cdot\frac{360}{\Delta\theta}+j+\frac{360}{\Delta\theta}-1\right),\\ where\,i=[1,N_{slice}-1],j=\frac{360}{\Delta\theta}\end{cases}\tag{14}$$

where $Vertex[\bullet]$ represents the spatial coordinates (x, y, z) of the mesh vertices; $f_{i,j}[\rho]$ and $f_{i,j}[\theta]$ denote the radius and angle of the j-th feature of the i-th slice, respectively; $Traingles(v_1,v_2,v_3)$ signifies the triangles of the mesh modeling, and v_1, v_2, v_3 are the serial numbers of the three vertices used to create the triangles. As a result, the geometric digital twin is constructed using geometric features extracted from point cloud data. Upon conducting periodic or planned inspections, the geometric digital twin is dynamically updated to facilitate condition assessment during the lifecycle service.

Algorithm 2: Extracting features for geometric digital twin of a pipeline

Input: All slices $S_{cal,i}=\{s_{cal,i_j}=(x_{i_j}-x_{c,i},y_{i_j}-y_{c,i})\in R^2, j=[1,N_r]\}, i=[1,N_{slice}]$
Output: Features of the geometric digital twin
Begin
1. Compute $\boldsymbol{F}_1$ according to Algorithm 1;
2. $\boldsymbol{D}_{x,base}=\boldsymbol{D}_{x,1}$, $\boldsymbol{D}_{y,base}=\boldsymbol{D}_{y,1}$, $\boldsymbol{F}_{base}=\boldsymbol{F}_1$;
3. $i=2$, set x_{min}, x_{max}, y_{min} and y_{max}, and $\psi_{thres}=0.995$;
4. while $i\le N_{slice}$:
   Compute $\boldsymbol{D}_{x,i}$ and $\boldsymbol{D}_{y,i}$;
   Compute $\psi(\boldsymbol{D}_{x,i},\boldsymbol{D}_{x,base})$ and $\psi(\boldsymbol{D}_{y,i},\boldsymbol{D}_{y,base})$;
   if $\psi(\boldsymbol{D}_{x,i},\boldsymbol{D}_{x,base})\langle\psi_{thres}$ or $\psi(\boldsymbol{D}_{y,i},\boldsymbol{D}_{y,base})\langle\psi_{thres}$:
      Compute $\boldsymbol{F}_i$ according to Algorithm 1, $\boldsymbol{D}_{x,base}=\boldsymbol{D}_{x,i}$, $\boldsymbol{D}_{y,base}=\boldsymbol{D}_{y,i}$, and $\boldsymbol{F}_{base}=\boldsymbol{F}_i$;
   else: $\boldsymbol{F}_i=\boldsymbol{F}_{base}$;
   i++;
5. return $\left\{\left(x_{c,1},y_{c,1},z_{c,1}\right),\boldsymbol{F}_1\right\},\left\{\left(x_{c,2},y_{c,2},z_{c,2}\right),\boldsymbol{F}_2\right\},\cdots,\left\{\left(x_{c,N_{slice}},y_{c,N_{slice}},z_{c,N_{slice}}\right),\boldsymbol{F}_{N_{slice}}\right\}$

2.3.3. Geometric digital twin-informed condition assessment

Cross-sectional deformation and blockage are crucial indictors in structural deterioration and functional degradation of the pipeline, respectively. The current Chinese code (CJJ 181–2012, 2012), and American standard (ASTM F-1216–22, 2022), and several studies (Bai and Bai, 2005; Hawari et al., 2018; Xu et al., 1998) have adopted the ovality indicator to measure pipeline deformation, but the ellipse fitting is not consistent with the actual situation (as shown in Fig. 1) and is also not accurate enough based on 2D images. Leveraging geometric digital twin, this paper presents a method for quantitative assessment of pipeline deformation and blockage, including ovality indicator (*OI*) of cross-section and effective flow volume reduction rate (*VI*), as demonstrated in Eqs. (15) and (16):

$$OI=\frac{D_{max,i}-D_{min,i}}{D_{base,i}}\cdot100\%\tag{15}$$

$$VI=\frac{V_{base}-V_{curr}}{V_{base}}\cdot100\%\tag{16}$$

where $D_{base,i}$ denotes the design inside diameter of the pipeline; $D_{max,i}$ is the maximum diameter from the i-th slice of the current model, while $D_{min,i}$ is that of the smallest; V_{base} is the effective flow volume of the design inside diameter, while V_{curr} is that of the current model. The mesh volume of the current model can be calculated using the VTK package (version: 9.2.6) in Python.

Leveraging the geometric digital twin, diameter D_{i_k} within a slice ($\left\{\left(x_{c,i},y_{c,i},z_{c,i}\right),\boldsymbol{F}_i\right\}$; $\boldsymbol{F}_i=\{f_{i_k},k=[1,360/\Delta\theta]\}$) can be calculated by summing the radius of the feature f_{i_k} with angle $(k-1/2)\Delta\theta$ and the radius of feature with angle $(k-1/2)\Delta\theta+180$. To enable quantitative identification and accurate classification of ovality and blockage in pipelines, this paper presents Eq. (17) based on *OI*, *VI* and the gradients

of diameter variations within a slice S_i.

$$\begin{cases} S_i \rightarrow Ovality\ Defect\ if \left|\dfrac{D_{i_k+1} - D_{i_k}}{\Delta\theta \cdot D_{i_k}} \cdot 100\%\right| \leq G_{thres}\ and\ OI > OI_{thres} \\ S_i \rightarrow Blockage\ Defect\ if \left|\dfrac{D_{i_k+1} - D_{i_k}}{\Delta\theta \cdot D_{i_k}} \cdot 100\%\right| \rangle G_{thres}\ and\ VI > VI_{thres} \end{cases} \tag{17}$$

where OI_{thres} and VI_{thres} are the thresholds for overall deformation and blockage respectively, which can be set to 2 % and 15 % correspondingly based on the standard (DNS-OS-F101, 2012; Munro et al., 2009; Rayhana et al., 2021) and specification (CJJ 181–2012, 2012). Once the threshold is reached, maintenance actions should be taken. D_{i_k+1} and D_{i_k} are the $(k + 1)$-th and k-th diameter of the i-th slice, respectively. G_{thres} is the threshold of the diameter variation per unit angle and can be set equal to OI_{thres}; it will increase significantly when there is blockage, whereas it changes slowly when there is only overall deformation.

3. Experimental verification

This section aims to validate the feasibility, accuracy and computational efficiency of the proposed methodology in pipeline deformation measurement and geometric digital twin modeling. The accuracy and computational efficiency of the deformation or profile measurement is fundamental to the geometric digital twin of the pipeline. The deformation experiment and accuracy estimation method are first designed, followed by a comparison with three widely-used point cloud processing algorithms to demonstrate the superiority of our method. Finally, the results of geometric digital twin-informed quantitative condition assessment are analyzed.

Table 1
Summary of deformations and sampling resolutions.

Case number	deformation*	Sampling resolution**	Scanning time	Points***
Case 1	Baseline (0 m)	1.534 mm	01 h:56 m:00 s	7,623,529
1–1	0.019 *m*	1.534 mm	01 h:56 m:00 s	7,356,234
1–2	0.038 *m*	1.534 mm	01 h:56 m:00 s	7,205,377
1–3	0.057 *m*	1.534 mm	01 h:56 m:00 s	7,666,036
Case 2	Baseline (0 m)	6.136 mm	00 h:30 m:34 s	1,021,139
2–1	0.019 *m*	6.136 mm	00 h:30 m:34 s	801,972
2–2	0.038 *m*	6.136 mm	00 h:30 m:34 s	828,698
2–3	0.057 *m*	6.136 mm	00 h:30 m:34 s	7,22,145
Case 3	Baseline (0 m)	12.272 mm	00 h:09 m:06 s	210,203
3–1	0.017 *m*	12.272 mm	00 h:09 m:06 s	220,350
3–2	0.036 *m*	12.272 mm	00 h:09 m:06 s	232,660
3–3	0.055 *m*	12.272 mm	00 h:09 m:06 s	175,210
Blockage	0 *m*	12.272 mm	00 h:09 m:06 s	129,053

* Note: the value of deformation is the maximum displacement change in six groups of target points (1–7, 2–8, 3–9, 4–10, 5–11 and 6–12).
** Note: the value of sampling resolution is the distance between two data points of the collected point cloud at 10 m.
*** Note: Points is the count of points contained in each slice of the point cloud.

3.1. Implementation details

Laboratory experiments were conducted for a total of thirteen cases, as shown in Table 1. The first twelve groups were deformation tests, which were carried out by spring compression to simulate the process of vertical loads on underground pipelines (Fig. 11c), followed by blockage test, which used irregular pieces of paper to simulate the appearance of gravel obstacles at the bottom of sewage pipes. The experiment was designed to verify the accuracy of deformation measurement rather than the mechanical properties of a pipeline. A plexiglass pipe segment with better deformation capability and ease of operation was employed in the experiments. To enhance the reflective ability of the plexiglass pipe surface and to better simulate the negative effects of rough concrete pipe surface, a layer of rough and opaque wallpaper was pasted on the inner surface of the pipe. The accuracy of the proposed method can be analyzed by comparing the targets of the total station measurement with the marks in the point clouds, as total station measurements are more precise than the raw point clouds. FARO Focus-3D and Leica total station were utilized, with technical parameters listed in Table 2. As shown in Fig. 11a and b, twelve total station target points were uniformly arranged on the inner surface of the pipe, with corresponding positions marked on the pipe surface for angles (e.g., $\theta^0_{c,ij}$ and $\theta^1_{c,ij}$) and radii (e.g., $L^0_{c,i}$, $L^1_{c,i}$, $L^0_{c,j}$ and $L^1_{c,j}$) in the point clouds. However, the accuracy should not be estimated by directly comparing the distances $L^0_{TS,ij}$ and $L^0_{PC,ij}$, because the distances between the target points and those between the marker points may not be equal. In this study, the indicators, as shown in Fig. 11c, were defined by Eqs. (18), (19) and (20) to assess the accuracy of point cloud data processing.

$$\begin{aligned} L^0_{TS,ij} &= \sqrt{\left(L^0_{o,i}\right)^2 + \left(L^0_{o,j}\right)^2 - 2\left(L^0_{o,i}\right)\left(L^0_{o,j}\right)\cos\left(\theta^0_{o,ij}\right)} \\ L^1_{TS,ij} &= \sqrt{\left(L^1_{o,i}\right)^2 + \left(L^1_{o,j}\right)^2 - 2\left(L^1_{o,i}\right)\left(L^1_{o,j}\right)\cos\left(\theta^1_{o,ij}\right)} \\ \Delta L_{TS,ij} &= \left|L^1_{TS,ij} - L^0_{TS,ij}\right| \end{aligned} \tag{18}$$

$$\begin{aligned} L^0_{PC,ij} &= \sqrt{\left(L^0_{c,i}\right)^2 + \left(L^0_{c,j}\right)^2 - 2\left(L^0_{c,i}\right)\left(L^0_{c,j}\right)\cos\left(\theta^0_{c,ij}\right)} \\ L^1_{PC,ij} &= \sqrt{\left(L^1_{c,i}\right)^2 + \left(L^1_{c,j}\right)^2 - 2\left(L^1_{c,i}\right)\left(L^1_{c,j}\right)\cos\left(\theta^1_{c,ij}\right)} \\ \Delta L_{PC,ij} &= \left|L^1_{PC,ij} - L^0_{PC,ij}\right| \end{aligned} \tag{19}$$

$$error_{ij} = \left|\Delta L_{PC,ij} - \Delta L_{TS,ij}\right| \tag{20}$$

where $error_{ij}$ denotes the error between the i-th marker and j-th marker in processing result of point cloud data; $\Delta L_{PC,ij}$ and $\Delta L_{TS,ij}$ are the deformation values obtained from the point cloud and total station, respectively; $L^0_{PC,ij}$, $L^1_{PC,ij}$ and $L^0_{TS,ij}$, $L^1_{TS,ij}$ are the distances between the i-th and j-th markers of the point cloud and total station before and after deformation; O represents the position of the total station, while C refers to the center of the point cloud slice; $i0$ ($j0$) and $i1$ ($j1$) denote the position of the i-th (j-th) marker point before and after deformation.

For algorithm parameter selection, the proposed methodology is robust, with critical input parameters only including Δt, $\Delta\theta$, Δx and Δy. Δt, the division step of the slice, takes into account the data distribution quality of the slice and the longitudinal resolution of the geometric digital twin. With pipeline defects (e.g., spalling and blockages) within a centimeter range not commonly considered, $\Delta t = 10\text{mm}$ is selected to ensure sufficient data in a slice for statistical distribution. $\Delta\theta$, the division step of the cell, is determined based on circumferential resolution and cell data distribution quality. $\Delta\theta = 360{\cdot}3/(786{\cdot}\pi) \approx 0.5^{\circ}$ is chosen. To accurately determine the similarity between slices for successful defect capture during region growing, both Δx and Δy are set equal to the scanner measurement standard deviation of 2 mm. All input parameters used in the proposed algorithm are listed in Table 3.

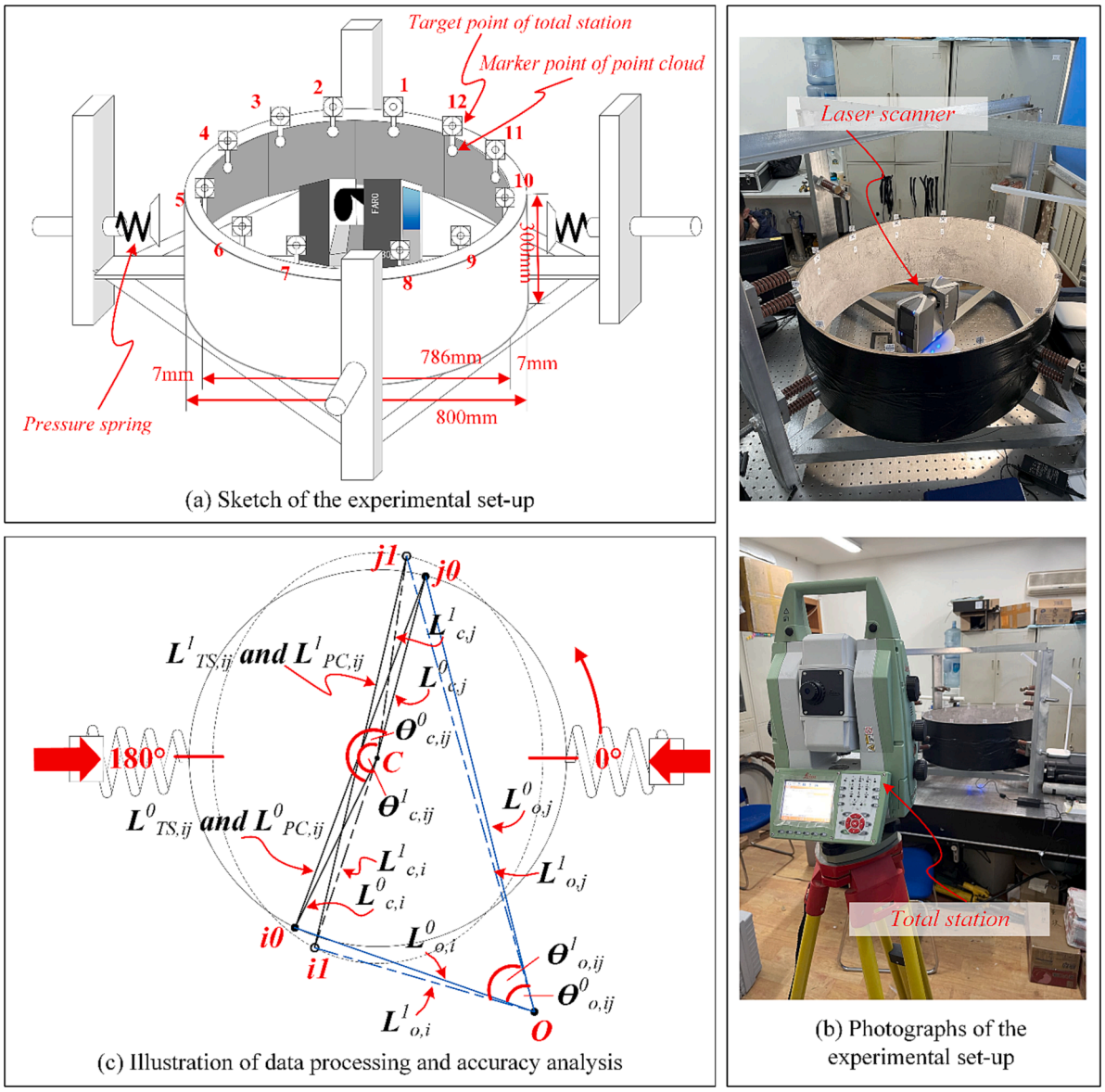


Fig. 11. Principle scheme of the deformation experiment.

Table 2
Technical parameters of the total station and laser scanner used in this study.

Equipment	Type	Data format	Angular accuracy	Distance accuracy
FARO Focus^s 70	Laser scanner	Point cloud with a large amount of noise and outliers	19″	Standard deviation of the 3D measurement at 10 m is 2 mm, and at 25 m is 3.5 mm
Leica FlexLine TS07	Total station	An exact number	0.5″	1 mm + 1.5 ppm, where 1.5 ppm is the uncertainty that needs to be increased by 1.5 mm per kilometer

Table 3
Input parameters used by the proposed algorithm.

Name	Notation	Values	Factors of selection
Division step of the slice	Δt	10 mm	Resolution of the geometric digital twin along the longitudinal direction; Quality of the slice's data distribution.
Division step of the cell	$\Delta\theta$	0.5°	Resolution of the geometric digital twin along the circumferential direction; Quality of the cell's data distribution.
Grouped interval size	$\Delta x(\Delta y)$	2 mm	Scanner measurement standard deviation.
Predefined slice size threshold	n_{r_min}	0.9 N_r	Number of points in a full slice (N_r) of the actual point cloud.
Probability threshold of inliers	P_{thres}	70 %	The ratio of inliers should be at least 50 %; Small value may increase iterations and computational time; large value may cause datasets containing more outliers to not converge in the algorithm; 70 % was set as an empirical reference value.
Predefined similarity threshold	ψ_{thres}	0.995	Obtained from actual similarity comparisons and experience.

3.2. Results

According to the input parameters in Table 3 and the experiments designed in Section 3.1, three widely-used point cloud processing methods, i.e., kernel density estimation (KDE), statistical-based outlier removal (SOR) and voxel grid-based sampling (Voxel), are compared

with our proposed methodology in terms of accuracy and computation time to demonstrate its superiority. Additionally, this section explores the feasibility of the geometric digital twin-informed quantitative assessment of pipeline deformation and blockage defect.

3.2.1. Accuracy evaluation

To analyze the performance of the four methods across various pipeline deformations and point cloud sampling resolutions, the laser scanner was employed to measure three different deformations at each of the three different sampling resolutions (as shown in Table 1). Next, errors were calculated using the four methods for six groups of target points (including 1–7, 2–8, 3–9, 4–10, 5–11 and 6–12), resulting in $3 \times 3 \times 4 \times 6 = 216$ error samples. Fig. 12 depicts the calculation results of all errors for the four methods. It is evident that the KDE method and the proposed method outperformed others, with their mean errors both below the 1 mm accuracy threshold, thus satisfying the requirements of deformation measurement. The SOR method demonstrated an accuracy of 1.1 mm, which followed the KDE and proposed methods. The Voxel method exhibited the worst performance, with an error of up to 1.52 cm. However, the maximum error of the SOR method approached 6 mm, significantly higher than that of the KDE and the proposed method, indicating that the proposed method is more robust than the SOR method. Moreover, the calculation results of geometric features shown in Fig. 13 further demonstrate the superior performance of KDE and the proposed method. Notably, regarding sampling resolution, the error of the proposed method, the KDE method and the SOR method all experienced a minor increase with lower sampling resolution. Conversely, the Voxel method displayed a different trend due to its poor outlier removal performance, as a point cloud with higher sampling resolution might contain more outliers. Although the KDE method demonstrated a more stable performance in comparison with the proposed method, this obtains at the expense of computational time. The computational efficiency of the four methods will be discussed below, which can reflect the potential of the proposed method to be applied to large-scale underground pipe networks.

3.2.2. Computational time

The aforementioned results demonstrate that both the KDE method and our proposed method can extract the geometric features of the pipeline surface point cloud with high accuracy, exhibiting a mean error of less than 1 mm. However, laser scanning point cloud data typically consists of millions of points, resulting in processing efficiency or computational time being one of the most crucial issues in assessing the practical application value of methods. Fig. 14 depicts the computational time required for extracting the geometric features of a slice for each of the four methods with different data volumes. From the perspective of processing efficiency, the performance of our method is comparable to that of the SOR method, with processing times of 171.73 s and 105.31 s at the 1.02 million points level, respectively. However, the KDE method becomes increasingly time-consuming as the volume of

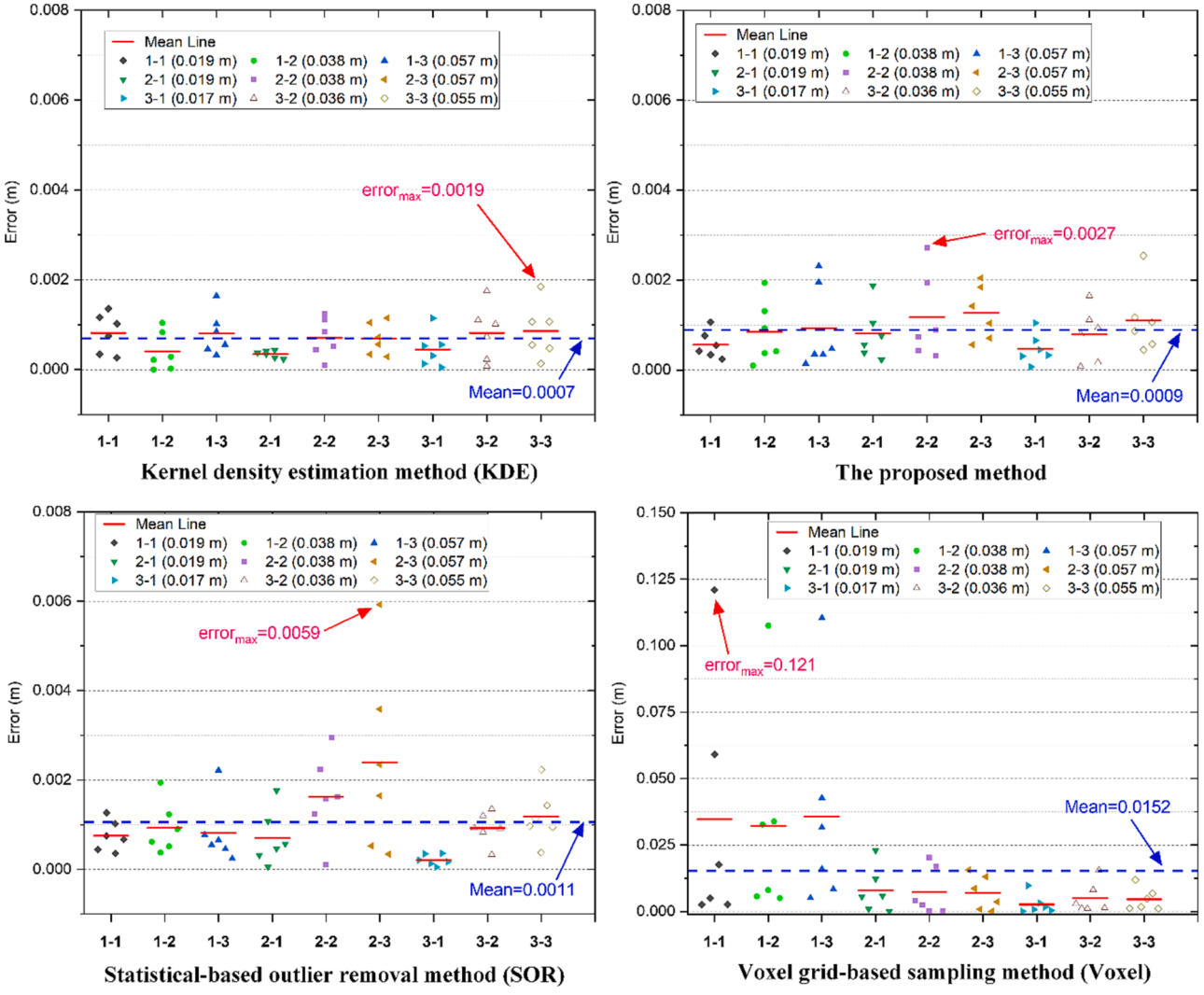


Fig. 12. The results of the errors using the KDE, SOR, Voxel, and the proposed method.

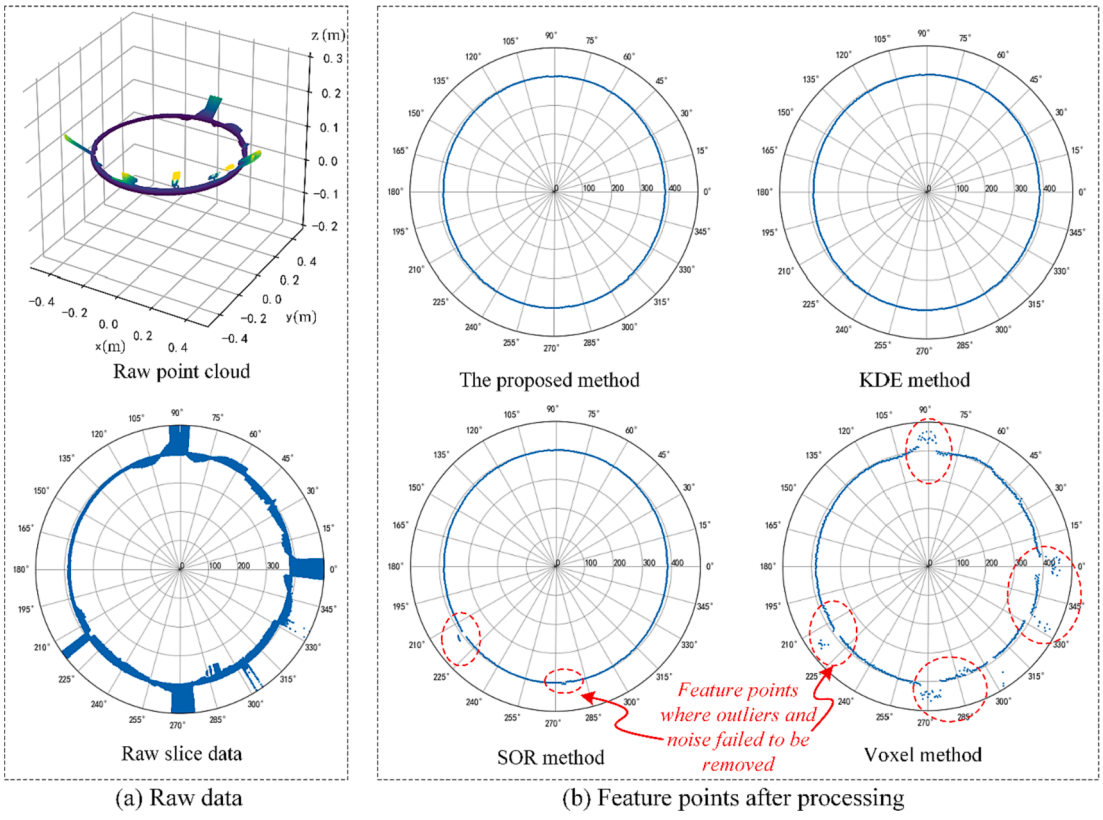


Fig. 13. Example of Case 1–1: (a) raw data; (b) performance comparison in computing feature points between the proposed method, KDE, SOR and Voxel.

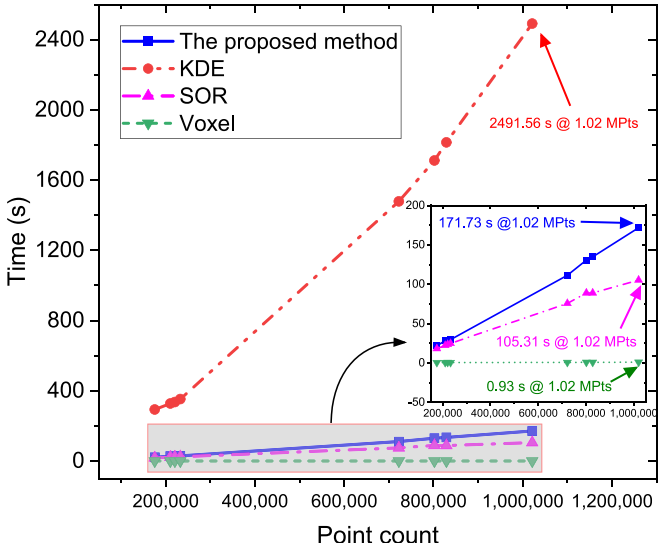


Fig. 14. A comparison of the computational time between the four methods.

data increase, reaching a processing speed of 2491.56 s at 1.02 million points. The KDE method relies on computing the contribution (weights) of each data point to the probability density (Picos et al., 2020; Silverman, 1986), and the performance of KDE methods is heavily influenced by the bandwidth selection (Armesto et al., 2010), leading to a challenge for processing large datasets. Although some techniques (such as k-fold cross validation) were developed to automatically determine bandwidth (Shimazaki and Shinomoto, 2010), they require a significant amount of run time, especially within dense laser scanning point clouds (Laefer and Truong-Hong, 2017). Compared with KDE method, the proposed method has the advantage of much smaller computing load. The method first removed outliers by clustering the data via DBSCAN which has a much smaller computational time than KDE. Then, in contrast to KDE that needs to compute the contribution of all point data (including outliers and noise), our method used a simple Gaussian fit to compute the contribution of the remaining measurement noise to the probability density, which is a process that has a small computational complexity and almost negligible computational time. The results in Fig. 14 showed that the proposed method achieves the same high accuracy as the KDE method but has a running time of only 7 % of that of the KDE method, indicating that the proposed method significantly outperforms the KDE method.

Finally, three representative examples using the proposed method are listed in Table 4 to illustrate the total computational time required for the complete processing workflow, from the raw point cloud to the geometric digital twin. It is evident that the majority of the computational time is spent on the DBSCAN-based clustering and fitting step. The running time of Point Cloud 2 is greater than that of Point Cloud 1 due to the higher volume of data that needs to be analyzed at each cell in a slice with high sampling resolution. Moreover, Point Cloud 3, which contains pipe defects, results in more slices that do not meet the similarity threshold, leading to a longer executing time compared to Point Cloud 1 (233.65 s for Point Cloud 3 vs. 52.71 s for Point Cloud 1). Consequently, it can be argued that the processing time primarily depends on not only the sampling resolution but also the presence of pipeline defects. In summary, the proposed method can process large datasets accurately and efficiently, with total processing times of approximately 52.71 s for 2.98 million points in Point Cloud 1, 115.53 s for 11.94 million points in Point Cloud 2, and 233.65 s for 3.74 million points in Point Cloud 3. Note that the performance reported herein was based on the implementation of all methods in Python 3.10 and was performed on a normal personal computer with the following system configuration: a 12th Gen Intel(R) Core(TM) i7-12700F CPU @ 2.1 GHz with 64 GB RAM, a GeForce RTX 3080 GPU.

It should be pointed out that the experimental tests were only

Table 4
Total computational time of the complete processing workflow.

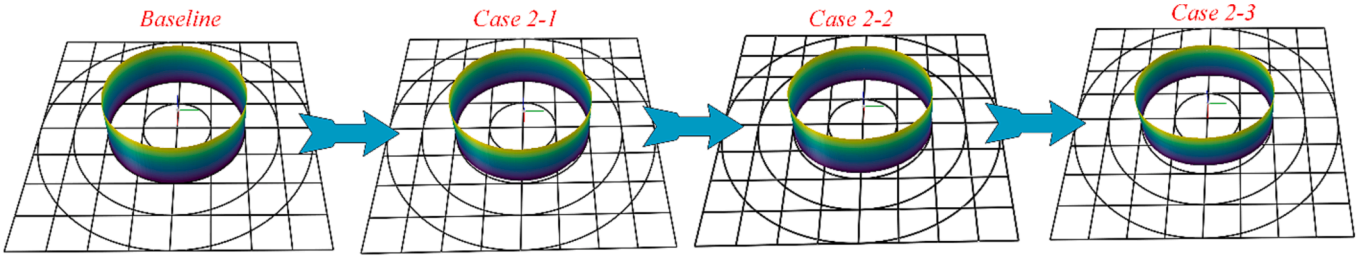
Point Cloud	Sampling resolution*	Description of the raw point cloud	Step						Total time
			1. 3D point-To-2D region		2. Geometric feature extraction		3. Geometric digital twin modeling		
1	12. 272 mm	Points: 2,987,356	Pipeline axis alignment	time: 0.26 s	LMedS-based slice center extraction	time: 4.55 s	Geometric features using DRG	time: 17.31 s	52.71 s
		Defects: No	Slice region subdivision	time: 0.83 s	DBSCAN-based clustering and fitting	slice**: 3 time: 26.61 s	Mesh modeling	time: 3.15 s	
2	6.136 mm	Points: 11,941,406	Pipeline axis alignment	time: 0.35 s	LMedS-based slice center extraction	time: 6.72 s	Geometric features using DRG	time: 35.69 s	115.53 s
		Defects: No	Slice region subdivision	time: 2.97 s	DBSCAN-based clustering and fitting	slice**: 2 time: 66.64 s	Mesh modeling	time: 3.16 s	
3	12.272 mm	Points: 3,742,547	Pipeline axis alignment	time: 0.38 s	LMedS-based slice center extraction	time: 5.4 s	Geometric features using DRG	time: 18.04 s	233.65 s
		Defects: Yes	Slice region subdivision	time: 0.83 s	DBSCAN-based clustering and fitting	slice**: 14 time: 205.56 s	Mesh modeling	time: 3.44 s	

* Note: the value of sampling resolution is the distance between two data points of the collected point cloud at 10 m.
** Note: the number of slices whose similarity does not reach the threshold.

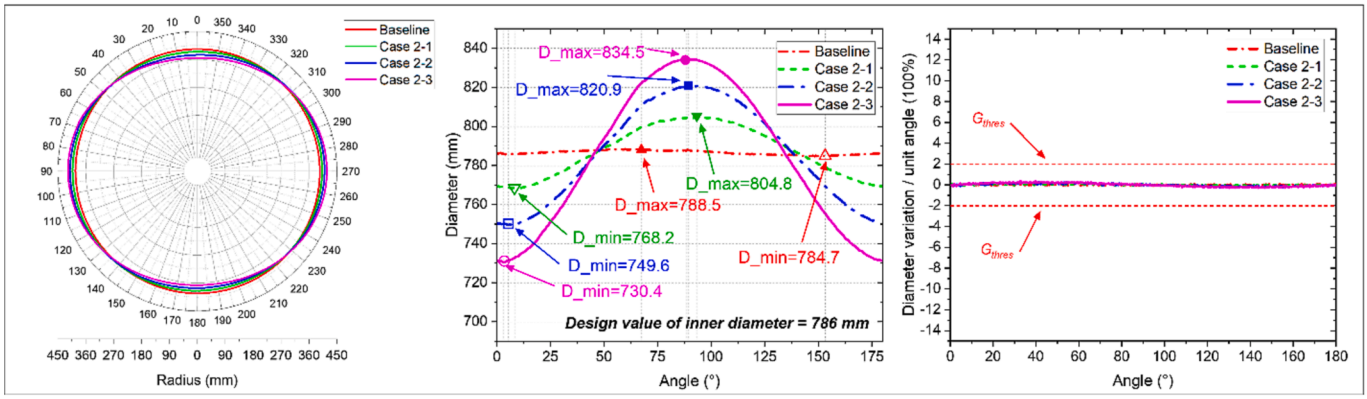
conducted on a limited length of pipe segment model, and the KDE method required a long computational time. The low computational efficiency of the KDE method would significantly hinder its application in practical large-scale pipe networks. In summary, the proposed method not only robustly achieves high accuracy but also maintains high computational efficiency, making it promising for pipeline engineering applications.

3.2.3. *Deformation and blockage assessment*

Underground pipelines, due to their buried nature and the harsh environment, presents challenges for inspectors to efficiently locate and quantify deformation and blockages. This makes it difficult to assess the level of aging and defects of pipelines in a timely manner. Geometric digital twin can facilitate comprehensive perception of pipe aging evolution and defect development in harsh subsurface environments through periodic monitoring of underground assets using laser scanner



(a) An example of pipe deformation evolution



(b) Diameter variations at the 25 mm cross-section

Fig. 15. Workflow for quantitative assessment of pipe ovality defects.

robotic systems (FARO, 2023). This section investigates the feasibility of geometric digital twin-informed quantitative assessment for deformation and blockages in pipelines.

Cases 2, 2–1, 2–2, and 2–3 in Section 3.2.1 were utilized as examples of various pipe deformations (Fig. 15a) to illustrate that the proposed method can accurately quantify pipe deformation through the geometric digital twin, providing a basis for structural safety assessment. Fig. 15b depicts the workflow of the proposed method to assess ovality defects in each cross-section of the pipe along the longitudinal direction. First, the trend of diameter variations of each cross-section should be analyzed according to Eq. (17) to avoid the interference from blockages. For instance, the cross-section at 25 mm was investigated to analyze its overall deformation. According to the deformation experiment scheme presented in Fig. 11c, the pipe was compressed in the angular directions of 0°-180° to simulate the vertical loads on the buried pipelines. The maximum and minimum diameters observed in the four tests were as follows: the maximum diameter of the baseline model on a 25 mm cross-section was 788.5 mm at an angle of 67.25°, and the minimum diameter was 784.7 mm at 153.25°; Case 2–1 had a maximum diameter of 804.8 mm at 93.25° and a minimum diameter of 768.2 mm at 8.25°; the maximum diameter of Case 2–2 was 820.9 mm at 89.25° and the minimum diameter was 749.6 mm at 5.25°; for Case 2–3, the maximum diameter was 834.5 mm at 88.25° and the minimum diameter was 730.4 mm at 3.25°.

The variations of diameter in the experiments were observed to exhibit a smooth wave-shaped curve before the structural failure of the pipeline, indicating that the pipe diameter did not change at a significant rate within a unit angle. The gradients of diameter variations demonstrated in the right diagram of Fig. 15b were found to be far below the 2 % threshold, further supporting the accuracy of the proposed method. The pipe *OI* results for the four tests are shown in Fig. 16. Taking the 25 mm cross-section as an example, the *OI* of the baseline model was 0.49 %, but the *OI* of Case 2–1, Case 2–2 and Case 2–3 all exceeded the 2 % threshold specified by the standard (DNS-OS-F101, 2012; Munro et al., 2009; Rayhana et al., 2021), with values of 4.65 %, 9.07 % and 13.24 %, respectively. Actually, in Case 2–1, the maximum variation of diameter was only 18.8 mm, emphasizing the importance of achieving millimeter-level accuracy for deformation measurement, which cannot be achieved by existing CCTV methods that rely on 2D image-based ellipse fitting.

While ovality can map the structural degradation of both water and sewer pipelines, the effective flow volume is another critical indicator reflecting the decline in pipe network function, particularly in sewer and rain pipelines. Blockages in sewer networks are the main cause of the reduction in effective flow volume, but quantifying this reduction is a challenge for municipalities and water authorities (Okwori et al., 2021). In this study, the feasibility of geometric digital twin-informed quantitative assessment for blockages using *VI* was demonstrated through a model test, as shown in Fig. 17a. Fig. 17b depicts the workflow of the proposed method for accurately locating and quantifying blockage using four representative cross-section examples located at different longitudinal positions of the pipe (i.e., −95 mm, 5 mm, 15 mm and 105 mm), respectively. As can be intuitively seen from the left and middle diagram of Fig. 17b, blockages were present in the pipe between approximately 125° and 170° of the circumferential direction, causing a sudden decrease in pipe diameter from 786 mm to between 700 mm and 740 mm. According to Eq. (17), the gradients of diameter variation were computed for each cross-section. As shown in the right diagram of Fig. 17b, all cross-sections exhibited gradients beyond the G_{thres} threshold, indicating that they should be included in the volume computation of the defect areas for *VI*. After calculation, the length of the blockage along the longitudinal direction of the pipe was determined as 290 mm, covering cross-sections from −145 mm to 145 mm.

The *VI* result of the pipeline was shown in Fig. 18. According to the design diameter in Fig. 11, the effective flow volume V_{base} was calculated as 140.71 L. The scenario with blockages in the pipe resulted in a decrease of volume to 129.30 L, representing an effective flow volume reduction rate of approximately 8.11 % calculated according to Eq. (16). According to Chinese code (CJJ 181–2012, 2012), the threshold value of VI is 15 %, therefore the pipe function was evaluated as still having adequate flow capacity. It is worth noting that the proposed method enables whole-process automated computation based on Python 3.10 programming. It achieves quantitative assessment with high accuracy and computational efficiency for deformation and blockage, which reprehensively map the structural and functional performance of pipelines, respectively, but have been challenging for existing methods. By leveraging the proposed method, pipeline operators and asset managers can more effectively monitor the condition of underground pipelines, and subsequently develop proactive maintenance plans. Ultimately, this approach can potentially reduce costs associated with pipeline failures and unplanned repairs, while also minimizing the environmental impact caused by pipeline leaks or road collapses. It should be noted that due to the limitation of laser scanning for underwater measurements, blockages below the flow line cannot be measured. In the future, 3D sonar data could be fused to overcome this limitation.

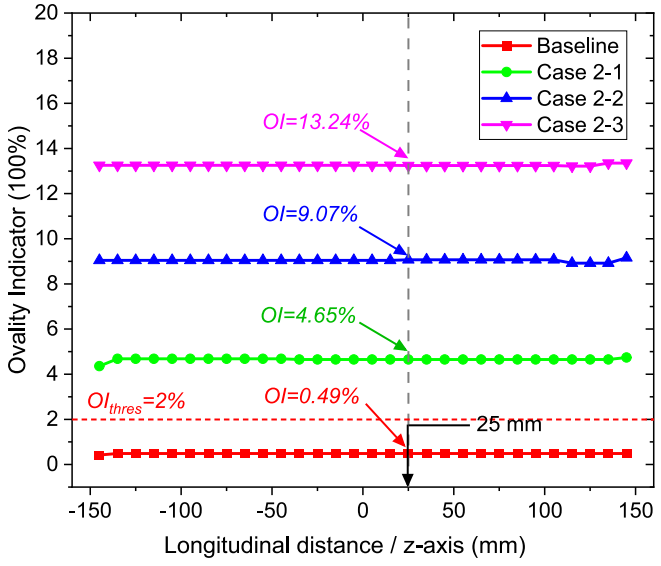


Fig. 16. The calculation results of *OI*.

4. Conclusions

The maintenance of underground water and sewer pipelines can be challenging due to their narrow internal environments and the presence of corrosive and toxic gases (Khan et al., 2021; Wang et al., 2021). To overcome these difficulties, this study proposes a novel method for fast and accurately processing point cloud data acquired by laser scanning technology to build a geometric digital twin of underground pipelines. The proposed method filters out noise and outliers based on the characteristics of laser scanning point cloud density distribution and finely describes the 3D surface of a pipeline to achieve an accurately mapping of pipeline conditions. A geometric digital twin-informed approach was presented to quantitatively assess the deformation and blockage in pipelines.

The accuracy and computational efficiency of the proposed method were demonstrated through experimental tests. Results indicated that the proposed method exhibited a mean error of less than 1 mm in deformation measurement, which satisfies the accuracy requirements for large diameter pipelines when compared to the measurements obtained the high-accuracy total station. Furthermore, the proposed method demonstrated superior accuracy and robustness when compared with the popular point cloud processing algorithms, namely SOR and Voxel methods. The computational time of the proposed method was

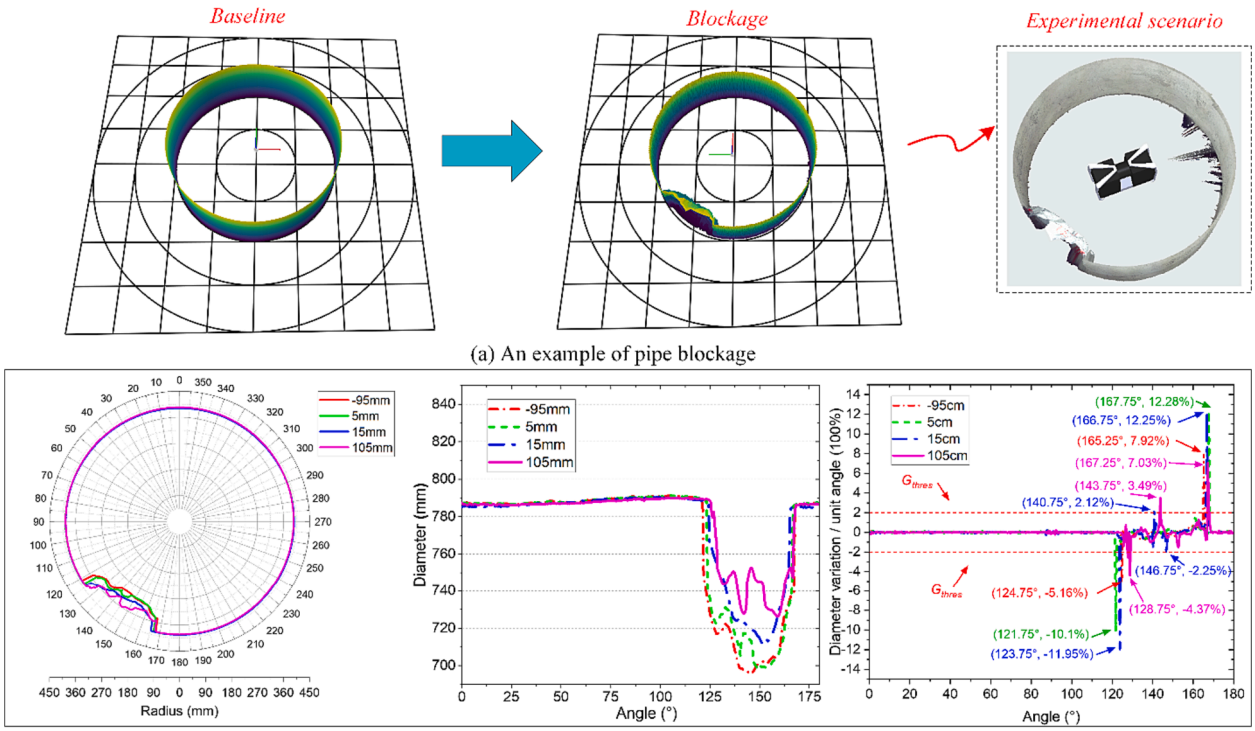


Fig. 17. Workflow for quantitative assessment of pipe blockages.

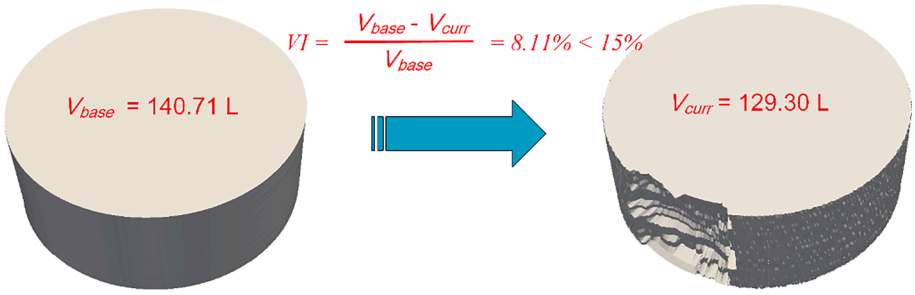


Fig. 18. Calculation results of *VI*.

only 7 % of that of the KDE method, indicating high computational efficiency of our method. Moreover, model tests demonstrated the proposed method enables quantitative assessment of pipeline deformation and blockage using geometric digital twin, which reveals promising applications in better informed O&M during life-cycle service. However, the method has some limitations at current stage, such as being unable to inspect the underwater part of the pipeline and not being suitable for bends with curved axes.

Further research directions include exploring the geometric digital twin and artificial intelligence (AI) for quantitative assessment of other defects, such as concrete spalling, corrosion, and open-joints. Additionally, an in-pipe inspection robot should be designed to carry the laser scanner for practical applications in engineering. Recently, robotics (RedZone Robotics, 2023) and portable 3D measurement systems (Ma et al., 2023b) are rapidly developing. In the future, we could further leverage these recent studies to validate the effectiveness of our method in dense point clouds acquired by these advanced devices and systems. Overall, the proposed method provides a promising tool in the assessment of underground pipe networks and can potentially reduce costs associated with pipeline failures and unplanned repairs, while also minimizing environmental impacts.

CRediT authorship contribution statement

Minghao Li: Methodology, Formal analysis, Software, Writing – original draft. **Xin Feng:** Supervision, Funding acquisition, Conceptualization, Investigation, Writing – review & editing. **Qunfang Hu:** Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Key R & D Program of China (Grant No. 2022YFC3801000) and the National Natural Science

Foundation of China (Grant No. 52079024).

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